

Crime Hotspot Prediction Using Geospatial Data Mining and Predictive Analysis

MASTER THESIS

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Abstract

Predicting where crime has happened or will happen has become a great concern for cities that are looking to move from reactive policing to a better informed and preventive urban safety planning. However, many hotspot prediction approaches still rely on historical crime data, which may show where crime has repeatedly occurred but often provide less information about the other conditions that shape and contribute to these patterns.

This study addresses that limitation by developing a geospatial machine learning model for predicting crime hotspots across the 77 community areas of Chicago.

The research adds historical crime data with spatially derived livability and accessibility indicators gotten from OpenStreetMap amenity data. Two complementary livability measures were developed for the analysis. First, the Unified Livability Index, this captures the density and diversity of key amenities within each community area. Second, the Distance-Based Livability Index which measures network-based accessibility to services using pedestrian street distances. These indicators were combined with temporal crime lag variables and a spatial lag feature derived from Queen contiguity relationships between neighbouring communities. Global Moran's was used to examine the spatial structure of crime, confirming that crime counts in Chicago show persistent and statistically significant spatial clustering between 2020 and 2025. This supported the inclusion of spatial dependence in the predictive modelling process. Logistic Regression, Random Forest, and XGBoost models were trained using historical data and evaluated on later years to reflect a realistic prediction setting.

Our results shows that the enhanced Random Forest model performed best overall, achieving stronger classification results than the models using only crime history. The analysis of feature importance confirmed that historical crime patterns remained the strongest predictors, but spatial lag and livability variables provided additional explanatory value. The combined use of ULI (Unifeid Livability Index), and DBLI (Distance Based Livability Index) was especially important, suggesting that amenity presence and actual accessibility capture different dimensions of urban context. These results showed that crime hotspot prediction can be improved when historical crime data are combined with spatial context and livability-based indicators.

However, the model showed difficulty in detecting emerging hotspots in 2025, showing the limitations of annually aggregated data for capturing sudden neighbourhood change. Overall, the research of this thesis contributes to GeoAI and spatial criminology by showing how open geospatial data, spatial statistics, and machine learning can be combined to produce more clearer and spatially grounded crime risk models for urban planning and public safety decision-making.

Keywords

Crime hotspot prediction; geospatial data mining; machine learning; spatial criminology; Chicago; livability index; accessibility; Random Forest; spatial autocorrelation; GeoAI.

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Science Pledge

I hereby declare that this thesis is the result of my own independent work. I have cited all sources and resources used in this thesis, and I have indicated their origin. This thesis was not previously presented to another examination board and has not yet been published. The GPT-5.1 model from OpenAI's ChatGPT was used as an aid to correct syntax errors in the code and also to suggest improvements. This was used sparingly and sensibly.

Ekata Leo Oni



04 May 2026

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Chapter 1: Introduction

1.1 Motivation and Background

All around the world, crime is posing a problem, which is putting police forces and security outfits under pressure. Security is a fundamental necessity for everyone, and this is why crime in urban areas continues to present a growing challenge for every stakeholder. To effectively predict crime is not just a necessity but, at this moment, a top priority for every stakeholder involved. Traditional hotspot prediction has mostly relied on the data of crimes that happened in the past to identify places where crime would likely to occur. This approach is useful for detecting spatial concentrations of crime, but on its own it does not fully account for the more broader, social and economic conditions that may influence those patterns (Eck et al., 2005; Chainey & Ratcliffe, 2013).

Correctly predicting crime or crime hotspots is producing more attention in a city like Chicago where diversity in the demography and disparities in socio amenities meet in an urban landscape which is quite complex. Understanding where crime occurs and the reason it does is important for creating or modifying existing policies for effective prevention. The traditional crime analysis methods which is focused on mapping hotspot manually or descriptive spatial statistics, have provided great insights into crime patterns around the world. However, these approaches are limited in their analytical depth, as they more often rely on historical crime records without considering other environmental factors that may likely influence criminal behaviour.

There have been improvements over the past decade because of advances in geographic information systems, spatial data mining, and machine learning which have enabled GIS analysts or experts to add or integrate a much wider range of spatial and socio-economic indicators into predictive frameworks (Chainey & Ratcliffe, 2013; Caplan et al., 2011). Such advances have made it possible to switch to proactive and data-driven prediction, rather than reactive crime mapping.

Moreover, accessibility modelling has become easier due to the recent advances in technologies associated with GIS, such as OpenStreetMap and spatial analytics libraries such as PySAL, etc. This is particularly applicable to services and facilities like health services, education, parks, transport and job opportunities. Availability of these resources has been identified as a significant aspect of livability and well-being (Pacione, 2003; OECD, 2020; U.S. EPA, 2021).

Recently, the reviews of the studies of the crime prediction also indicate that such contextual geospatial variables may enhance predictive models in case they are used in combination with crime history as opposed to when they are seen as independent issues (Du and Ding, 2023; Kounadi et al., 2020).

In this thesis, one can see an extensive study on how incorporating data on livability such as distance to social amenities such as parks, hospitals and other social amenities together with historical crime data could be a better predictor of crime.

Chicago is highly suited to study crime prediction due to its open crime data as provided by the Chicago data portal. The city offers open crime data in detail back to 2001 and in addition, with boundary data and other spatial resources to aid the urban analysis. Nonetheless, predictive studies continue to rely primarily on the past, and less on the overall socio-spatial context. This thesis responds to that gap by integrating historical crime data with geospatially derived indicators of accessibility and livability in order to develop a more spatially informed approach to crime hotspot prediction.

In summary, this thesis is aimed at looking at the feasibility of adding livability factors to historical crime data to show its effectiveness. By testing and validating several machine learning methods to see which one is best at combining the data for better prediction. Thereby offering recommendations and effective strategy

1.2 Problem Statement

The purpose of this thesis is to address the challenge of combining spatial analysis with machine learning to develop a model that integrates both historical crime data and livability indicators that are spatially derived. Crime remains a significant challenge, and current research suggests that the prediction of crime hotspots relies more on the spatial and temporal distribution of historical incidents, without considering the socio-economic and environmental aspects of criminal behaviour (Almanie et al., 2015; Chainey et al., 2008). Although machine learning models such as random forest and support vector machines have improved the accuracy of prediction (Zhang et al., 2020; Butt et al., 2021), these models see spatial data as tabular inputs, and as a result of this neglecting spatial dependence and context variability (Deng et al., 2023). The result is that predictions may lack interpretability and fail to identify local enablers of crime, including accessibility to public amenities or socioeconomic factors (Lamari et al., 2020; Yue & Chen, 2024).

The areas such as this exist in Chicago where these factors are critical. South and west side communities of the city, such as those, experience more socio-economic disadvantages, less access to healthcare, low employment, and infrastructure, which might lead to high crime rates. Traditional hotspot detection methods, such as Kernel Density Estimation (KDE) and Getis-Ord Gi* mapping, show where crime clusters occur but not why they emerge. The challenge, therefore, lies in developing a predictive model that not only forecasts crime hotspots with improved accuracy but also integrates the socio-spatial context from which these patterns are derived.

1.3 Research Aim and Objectives

The purpose of this thesis is to create a geospatial data system which will integrate spatial accessibility and livability indicators along with machine learning model which will be used to better predict crime hotspots, of which the focused research area in this thesis is the city of Chicago. To be more specific, it will assess whether incorporating socio-economic and environmental variables derived from spatial data can really improve the accuracy and reasoning of crime prediction models.

To address the research aim, the following sets of research objectives were derived:

- 1 Acquire and integrate multiple spatial data, including historical crime records, socio-economic attributes or amenity locations within the 77 community areas of Chicago.
- 2 Construct and compare two livability models: a unified livability index (ULI), which will be based on normalized socio-spatial scores, and a distance-based livability index (DBLI), which will be derived from average distances to key amenities
- 3 Evaluate and then develop machine learning models such as Random Forest, Logistic Regression, and XGBoost for predicting crime hotspots using both historical and socio-economic datasets.
- 4 Apply spatial analysis techniques such as Kernel Density Estimation (KDE), spatial lag modelling, and geographically weighted regression (GWR) to explore spatial dependence and local variation in crime livability relationships.
- 5 Interpret the influence of spatially derived socio-economic features or variables on prediction accuracy and discuss their implications for data driven urban safety planning.

1.4 Research Questions

To achieve the research objectives, this study will address the following key research questions:

- 1 Does integrating livability indicators, specifically the unified livability index and the distance based livability index, along with historical crime variables, improve hotspot prediction when compared with models based only on crime history.
- 2 Which historical, spatial, and livability related variables contribute strongly to hotspot classification across 77 community areas of Chicago, and how does these variables reflects on the wider socio-spatial inequalities within the city.
- 3 Does explicitly incorporating spatial dependence and local variation, through tools such as spatial lag features, Moran's I, and Geographically Weighted Regression, improve both the predictive quality and the spatial plausibility of resulting hotspot maps when compared with non-spatial machine learning models.

1.5 The Study Area (City of Chicago)

In comparison with some other major cities in the United States, Chicago has a high rate of violent crime, which makes it unsafe to go out. Chicago has 77 community areas, each with its own demographic and socio-economic characteristics. The city has spatial inequality, especially across its north, west, and south sides, where differences in income, education, infrastructure, and service access remain pronounced. These inequalities are also visible in crime geography. Each community serves as a spatial unit of analysis. The city of Chicago was selected as it is a stable administrative area where crime and demographic statistics are reported.

The city of Chicago presents a strong research opportunity in the area of spatial crime patterns due to the availability of over twenty years of crime data in its open data portal that is organized by location, time, and type of offence. Such richness of publicly-available information makes the city very suitable to spatio-temporal analysis. Moreover, the overall spatial data environment of Chicago, comprising administrative areas, transit systems, and facility layers, facilitates the incorporation of contextual geospatial variables into predictive processes.

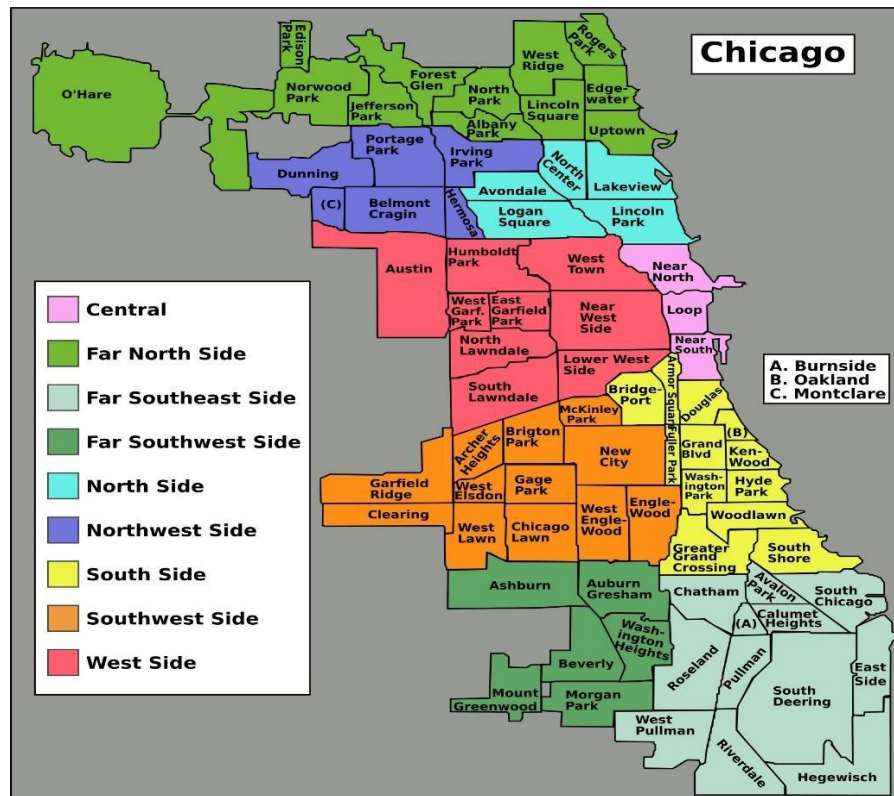


Figure 1. Map of Chicago Community Areas. From Chicago Community Areas Map, by P. Fitzgerald, n.d., Wikimedia Commons (<https://commons.wikimedia.org/w/index.php?curid=3902994>). CC BY-SA 3.0.)

Collectively, this is what renders Chicago an excellent study subject in a thesis that aims to relate crime prediction to the issues of accessibility, livability, and spatial inequality. The richness of the city in terms of data also enables the proposed framework to be formulated in such a manner that they can be extended to other cities.

1.6 Theoretical Framework: Spatial Criminology and Accessibility

The theoretical foundation of this research is spatial criminology, a discipline that is interested in the influence of place and opportunity to the behaviour crime. At the heart of this is the Routine Activity Theory (Cohen and Felson, 1979) that postulates that crime is the result of an interplay of three factors in time and space that include the presence of a motivated offender, an opportunity target and the lack of capable guardianship. The convergence is not accidental, but the result of where people travel, what they do and when they do it. This reasoning is extended by crime pattern theory, which demonstrates that offenders establish zones of awareness by their day to day travels, and that crime is concentrated around known nodes, routes and edges within the urban landscape (Brantingham and Brantingham, 1993). The

contribution made by Rossmo (2000) to geographic profiling further complicates this by showing that the distance attenuation, physical obstacles, and geographical landscape backcloth combine, having an effect on spatial distribution of the offense.

The relationship between these theories and the current study is the existence of urban livability as a measure of opportunity structure. Lack of access to healthcare, education or work does not directly lead to crime but does create the context including economic stress, social disorganisation and lessened guardianship that may make the community more vulnerable.

This study aims to bring these theoretical relationships into practice using operational tools in a geospatial machine learning model, where spatial dependency and environmental context are not put in the noise but become inputs of the model.

1.7 Scope and Limitations

This paper deals with crime hotspots forecasting in the 77 communities of Chicago, using the crime data of 2001-2025. Special focus is in on the more recent range of 2020-25, as an expression of the desire to reflect more up-to-date urban realities as opposed to be guided by trends that might no longer be valid. The main data sets will be the publicly available open crime data of Chicago and the OpenStreetMap amenity layers.

Several limitations are worth acknowledging upfront. Working at the community area level means that intra community variation in crime risk is smoothed over inevitably, two neighbourhoods within the same community area may have different profiles. The accessibility metrics are computed using representative centroid points for each community, which may not accurately capture the experience of residents living at the edges of a given area. There is also the question of OpenStreetMap data quality, which is known to vary considerably across cities and regions, and could introduce inconsistencies in the amenity layers used. Finally, the absence of individual level offender data constrains what can be said about crime mobility and offender decision making. These constraints are there and are real, but they do not undermine the broader value of the approach because the methodology is designed in a way that could scale to finer spatial resolutions if more granular data becomes available.

Chapter 2: Literature Review

Introduction

A combination of geospatial data mining and the appropriate machine learning model is excellent in detecting, forecasting and preventing crimes. Place and crime have been a topic of study among criminologists and urban scholars over decades but the instruments that can be used to analyze the relationship have evolved significantly. It is simply the person or the body that determines what is crime analysis. In the case of international practitioner organizations such as the International Association of Crime Analysts, it is concerned with the use of qualitative and quantitative techniques to assist the operational requirements of the police agencies and communities (IACA, 2014). The net is however attempted to be widened by academic definitions. Boba Santos (2012) places the crime analysis in a wider context of identifying and preventing patterns, whereas Ratcliffe (2016) directly relates it to the intelligence led strategies that define priorities in enforcing actions at the organizational scale. Older approaches were based on descriptive mapping to demonstrate crime location, more recent approaches have shifted towards spatially explicit computational approaches which utilize GIS, spatial statistics, and more recently machine learning.

This change has not come in a vacuum, but has been fueled by the change in theories over the years that provided the researchers with the incentive to not only take a careful look at where crime occurs, but also why.

Both routine activity theory (Cohen and Felson, 1979) and crime pattern theory (Brantingham and Brantingham, 1993) identify the urban environment as an influencing factor with criminal events attributed to the daily traffic of people in space. These bases have been developed by methodological contributions of crime mapping (Chainey and Ratcliffe, 2013), spatial dependence modelling (Anselin, 2003) and more recent syntheses of forecasting (Kounadi et al., 2020; Du and Ding, 2023), that have added to what can be measured, explained and predicted.

It is based on that path and is followed in this chapter. It starts with the criminological theories that render spatial modelling theoretically significant, followed by the development of analytical tools, goes on to view livability as a contextual issue in crime, and lastly ends by pointing out the gaps that led to this study.

2.1 Theoretical Foundations of Spatial Criminology

It is highly significant and worthy before we proceed to the methods that we should find out why crime should be expected to cluster at all in space. Such subsequent analytical decisions as the use of spatial weights, metrics of accessibility, and place level predictors will be justifiable in case there is some theoretical rationale as to why place matters. There are two bodies of theory which offer that justification.

2.1.1 Routine Activity Theory (RAT)

Routine Activity Theory of Cohen and Felson (1979) has been one of the most referenced theories and most powerful ones in spatial criminology. It provides a simple explanation of crime that does not rely on hypotheses regarding motivation of offenders. According to it, a crime is committed when three conditions are met at the same time and place, namely: a motivated criminal, an appropriate target, and the lack of adequate supervision. What the theory basically suggests is that instead of merely concentrating on the psychology of offenders we should also consider the situational factors that enable crime to occur. This then would prove very pertinent to spatial analysis as the characteristics of visibility, light, accessibility, and proximity to transport, etc., all form the local opportunity structures.

2.1.2 Environmental Criminology

This reasoning is then carried over by environmental criminology to the view that crime is a product of the interaction between routine movement and place characteristics. Crime pattern theory, specifically, suggests that criminals act in spaces of awareness that are developed by moving between network nodes like home, work and recreational locations that are linked by paths and bounded by edges (Brantingham and Brantingham, 1993). This contributes to the tendency of crime to be clustered around activity spaces that are known and common routes of use.

In applied work, this theoretical perspective has strongly influenced GIS-based crime mapping and hotspot analysis. Techniques such as Kernel Density Estimation and hotspot analysis are widely used for this purpose in applied crime mapping and intelligence-led analysis (Ratcliffe, 2010; Chainey & Ratcliffe, 2013). Additionally local spatial clustering measures such as the Getis-Ord G_i^* statistic provide a formal basis for identifying statistically significant hot and cold spots (Getis & Ord, 1992).

2.1.3 Journey-to-Crime Theory

Journey-to-Crime Theory adds a spatial behavioural dimension to crime analysis by emphasizing how offender's routine activity spaces and travel costs shape where crimes occur. Rossmo (2000), in the context of geographic profiling, proposed the idea that offenders crime locations tend to be spatially structured in relation to anchor points e.g., residence or activity nodes, consistent with distance-decay reasoning in which the likelihood of offending generally decreases with increasing distance. Although this study does not model individual offender journeys, the underlying logic supports using proximity and accessibility indicators at the community scale; areas with higher accessibility to potential targets e.g., commercial activity nodes, may face elevated risk, while differences in guardianship can further condition where opportunities concentrate (Rossmo, 2000; Cohen & Felson, 1979).

2.2 The Evolution of Crime Analysis and Prediction

The Evolution of Crime Analysis and Prediction Methods for studying crime have changed substantially over time. Crime analysis has not always looked the way it does today because each phase has reflected not only improved technologies but also changing assumptions about what crime analysis is meant to do, which is to describe, explain, or predict. This improvement is not driven just by any single breakthrough, but by the gradual improvement of better data, better tools, etc.

2.2.1 Pre-Computer Era: Manual and Descriptive Mapping

Before the widespread use of computers, for much of the twentieth century, police analysts worked with what they had. Crime mapping relied largely on manual methods such as pin maps, hand-drawn charts, and paper records that allowed them to mark where incidents occurred and spot rough concentrations (Ratcliffe, 2010). These approaches were limited because they were slow, labour intensive, and offered little capacity for temporal comparison or statistical testing. What they did do, however, is that they established an important empirical insight that would prove foundational, which is that crime is not spread evenly across a city but clusters together. Even in this early phase, analysts could see that some locations experienced repeated concentrations of incidents, an observation that later became central to hotspot research and GIS based spatial analysis (Ratcliffe, 2010; Chainey & Ratcliffe, 2013).

2.2.2 Computational Era: GIS and Spatial Statistics

The arrival of GIS tools in the 1980s and 1990s changed what analysts could do with incident data. GIS became increasingly central to policing and crime analysis. Rather than plotting points by hand, it enabled analysts to map incidents more systematically. Crime records could now be overlaid with street networks, census boundaries and land use data, and patterns could be viewed at various geographic scales with significantly greater accuracy (Chainey & Ratcliffe, 2013; Ratcliffe, 2010). Hotspot detection was an important component of this shift and approaches like Kernel Density Estimation and local cluster statistics provided more objective means of detecting concentrations of crime (Getis & Ord, 1992).

Simultaneously, spatial statistics attracted the interest to spatial autocorrelation, or the propensity of closely related places to exhibit similar values. This was what the pin map age had been feeling its way to. The I became a standard global measure of Moran to test whether patterns are not spatially random (Moran, 1950), and local statistics to detect the location of clustering (Getis and Ord, 1992). These developments highlighted a key methodological problem with conventional regression models that assume independent observations can be biased when spatial dependence is present. This led to wider adoption of spatially explicit models such as spatial lag/error frameworks (Anselin, 2003) and methods designed to capture spatial non-stationarity, including Geographically Weighted Regression (GWR) (Fotheringham et al., 2002).

2.2.3 Machine Learning Era: Pattern Recognition

Since the 2000s, advances in computing capacity and the availability, and access to large urban datasets have encouraged a shift from mainly explanatory analyses to predictive modelling. Random Forests are machine learning methods well suited to crime related data because they can capture non linear relationships and interactions without needing strict assumptions (Breiman, 2001). Studies have shown that machine learning baselines such as Random Forests and Support Vector Machines can perform well, and that performance can improve further when the models include non crime contextual information (Kang & Kang, 2017). However, recent reviews points out the limitations in many crime prediction studies, such as weak interpretability and insufficient incorporation of spatial temporal structure and contextual geospatial predictors (Kounadi et al., 2020; Du & Ding, 2023).

2.2.4 GeoAI Era: Spatial Intelligence

GeoAI, or Geospatial Artificial Intelligence, has become more common as a framework in GIScience that leverages advances in machine learning for geospatial data and spatial analysis (Li, 2020). In crime prediction, recent reviews indicates that adding geospatial features (e.g., accessibility and built-environment indicators) alongside crime history can improve understanding and modelling of spatial crime patterns (Du & Ding, 2023; Kounadi et al., 2020).

Another approach to crime forecasting approach is Risk Terrain Modelling (RTM) (Caplan et al., 2011), which operationalizes environmental risk factors as spatial layers to produce composite risk surfaces. More recent work uses graph-based deep learning to represent spatial units as connected nodes and learn spatio-temporal dependence patterns; for example, Hou et al. (2022) propose an attention-based graph model for urban crime prediction, and Sui et al. (2024) apply spatio-temporal graph attention models to street-level incident calls. These methods show the value of including accessibility and contextual geospatial indicators into predictive pipelines.

2.3 Urban Livability and the Spatial Determinants of Crime

The GeoAI approach suggests that context, and not only past events, helps predict crime. In this thesis, livability is defined as the combined availability and accessibility of key daily life resources such as healthcare, education, parks, employment opportunities, and public transport at the level of Chicago's community areas. It considers both the presence of amenities and how residents can easily reach them in their day to day routines.

The traditional frameworks see livability as a multidimensional system that combines environmental conditions, services, safety, and social context rather than a single indicator (Pacione, 2003; OECD, 2020). In applied index design, accessibility and walkability type measures provide an operational way to represent access to everyday opportunities and services (U.S. EPA, 2021). In the crime literature, CPTED emphasizes how the built environment and place management such as surveillance, access control, territoriality, and maintenance can shape opportunities for crime and perceived safety (Cozens & Love, 2015). At the neighborhood scale, collective efficacy has been repeatedly linked to lower violence, indicating that community level social processes are important factors connecting place conditions to crime outcomes (Sampson et al., 1997; Hipp & Wickes, 2017). Taken together, these perspectives justify using livability indices that include accessibility-related indicators as

contextual predictors or proxies for opportunity or guardianship and neighborhood functioning when modelling spatial crime patterns.

In addition to individual case studies, several frameworks define livability in a multi-dimensional way. The OECD well-being framework includes domains such as income, jobs, housing, health, and safety as core components of quality of life (OECD, 2020). Similarly, UN-Habitat's City Prosperity Initiative operationalizes a "quality of life" dimension including health, education, safety/security, and public space (UN-Habitat, 2016). At the neighbourhood level, Australian liveability research operationalises livability using GIS-based indicators of access to daily living destinations and walkable urban form (Lowe et al., 2013). Technical resources such as the U.S. EPA's National Walkability Index shows how urban form and access related variables can be converted into standardized, comparable indicators for spatial analysis (U.S. Environmental Protection Agency, 2021).

To capture these dynamics for Chicago, this thesis operationalizes livability by computing two complementary measures derived from OpenStreetMap (OSM) data:

- Unified Livability Index (ULI): Based on the density and diversity of important amenities within a community.
- The Distance-Based Livability Index (DBLI) is calculated as the mean network distance (accessibility) from a community's representative point to the nearest key amenities along the pedestrian street network.

By adding both indices, this research accounts for both the quantity and the accessibility of urban amenities, thereby producing a spatially grounded measure of community well being to test as a predictive feature.

2.4 Geospatial Data Mining and Predictive Modeling

The shift toward machine learning in crime prediction reflects a broader dissatisfaction with the limits of models that assume linearity and stationarity in relationships that are neither. The extraction of predictive patterns of spatially referenced data, with active consideration of the dependence structures that are considered a nuisance by conventional methods, is known as geospatial data mining as it is used in this thesis. Even when the main model is a machine learning algorithm, spatial diagnostics that tell us whether the residuals of your predictions themselves are spatially patterned can be useful, such as Moran I, local cluster statistics, specifications of spatial regressions (Anselin, 2003; Fotheringham et al., 2002).

A similar argument is made by recent reviews of the crime forecasting literature, which suggest that models solely based on past crime counts work well in predicting past locations of crime, but are less helpful in explaining why a neighbourhood is susceptible in the first place. Both Du and Ding (2023) and Kounadi et al. (2020) argue the necessity to include contextual geospatial predictors (built environment characteristics, accessibility proxies, socioeconomic indicators, and crime history) and transparent validation designs that would enable scrutinizing the results instead of accepting them as is.

- Random Forest (RF): The original definition by Breiman (2001) proved that when using large numbers of decision trees that are trained on bootstrap samples, with randomised feature selection at each split, an ensemble is formed that generalises better than any single decision tree, and far more resistant to overfitting. Importantly to this study, the Random Forest algorithm can deal with non-linear relationships and higher order interactions between predictors without the analyst having to pre-specify a parametric form. This becomes a significant strength in the context of a heterogeneous combination of crime history, accessibility indices and socioeconomic variables whose association with crime may not be straightforward and consistent across the 77 community areas of Chicago (Breiman, 2001; Pedregosa et al., 2011).
- Boosting (XGBoost): XGBoost is fast and efficient for gradient-boosted decision tree framework that builds an additive model by sequentially fitting trees to minimize a specified loss function, with strong emphasis on computational efficiency and scalability (Chen & Guestrin, 2016).
- Logistic Regression (LR): Logistic regression is included as a transparent baseline because its coefficients provide a direct indication of the direction and relative contribution of each feature. However, in its standard form it assumes a linear log-odds relationship and does not explicitly model spatial dependence unless spatial features such as lag terms are introduced (Pedregosa et al., 2011; Anselin, 2003).

Previous research provide a foundation for this approach. For example, Bogomolov et al. (2014) show that adding human mobility proxies derived from mobile data with demographic context can improve crime-related predictive modelling. More broadly, recent reviews of spatial and spatio-temporal crime forecasting emphasize that incorporating contextual geospatial features such as built environment, socio economic factors, and accessibility, in

addition to past crime data, can make models more useful and easier to interpret (Du & Ding, 2023; Kounadi et al., 2020).

2.5 Identified Research Gaps

Although there is an improvement in predictive models and accuracy, it has its limitations or gaps. The most intractable is the fact that historical counts of crime are treated as a predictor in and of themselves. Models based on this assumption may do very well on historical test sets, but they are engaging in a similar activity to pattern extrapolation rather than actual explanation.

Another issue of concern is interpretability. Gradient boosting models and, more recently, Graph Neural Networks have pushed the limits of classification accuracy to astounding levels, though the mechanisms of their predictions can be opaque. A model that fails to answer the question of what features it cares about, or in what way its predictions are least accurate can provide little help to planners or community organisations who attempt to make defensible decisions regarding resource allocation. The lack of interpretability is of little use in a policy context.

Lastly, there is the underutilization of open geographic data sources such as openstreetmap especially as a predictor of crime modelling. OSM provides high-resolution, constantly updated data about the location of schools, health facilities, parks, transport nodes and commercial services within entire cities, but little research has tried to deduce systematic, reproducible livability metrics using this data. Where OSM data is applied to crime research, it is applied as context mapping, and not as an element in prediction models.

Chapter 3: Methodology

Overview

This chapter sets out how the study was conducted like what data were collected, how they were processed, and what analytical decisions were made along the way. The approach is best described as a hybrid GeoAI framework. It combines spatial statistics, machine learning, and accessibility based livability measures, rather than treating any one of these as sufficient on its own. The reasoning behind this combination is developed throughout, but the short version is that crime prediction models which rely only on historical incident counts tend to reproduce past patterns without explaining them. Incorporating spatial context and environmental accessibility metrics is an attempt to do something more analytically useful.

The workflow goes through data acquisition, preprocessing, spatial integration, feature engineering, model development, evaluation, and spatial interpretation.

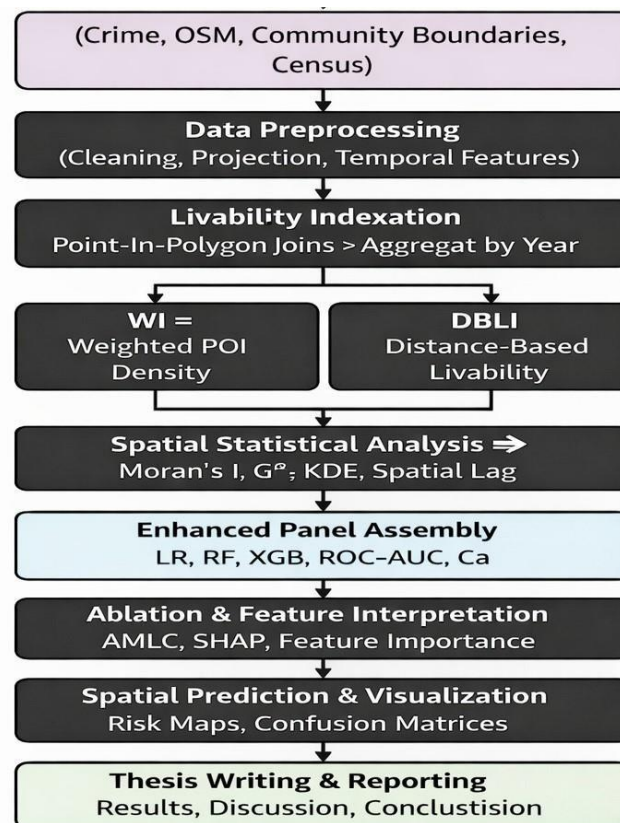


Figure 3.1 provides a visual overview of the workflow.

3.1 Study Area

Chicago has 77 community areas, fixed administrative boundaries that have not changed much since the 1920s and as such, they are best suited to be used as space units of analysis. Both the crime records as well as the demographic data are always reported at this level and this eliminates one of the most frequent headaches in urban spatial research that is the need to have to reconcile data gathered on the basis of different systems of boundaries.

The city is also a really fascinating subject of such a study. Its city form is very heterogeneous with severe socioeconomic differences between neighbourhoods which are geographically nearest to each other. It is that variation that makes the research have traction rather than be a complication.

In this study, all spatial operations were performed in EPSG:3435 projected coordinate system (NAD83 / Illinois StatePlane East) in order to get the correct distance measurements, and then the end product map reprojected to EPSG:4326 (WGS84) to be visualised.

3.2 Data Sources

The research relies on four primary sources of data, which are as follows in Table 3.1:

Table 3.1. Primary Datasets Used in the Study

Dataset	Source	Description	Temporal Coverage
Chicago Crime Data	City of Chicago Data Portal	Incident level records of reported crimes (location, type, date)	2020 - 2025
Community Area Boundaries	City of Chicago Data Portal	Polygon shapefile of 77 community areas.	Static
Amenity Data	OpenStreet Map (via OSMNX)	Point of interest: schools, hospitals, parks, transports etc.	2025
Livability Index (ULI and DBLI)	Computed in this study	Derived from amenity proximity and accessibility.	2025

Spatial weight matrix	Constructed via PySAL	Queen relationships communities	contiguity between	Static
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3.3 Data Preprocessing

3.3.1 Crime Data Cleaning and Aggregation

The raw crime data covers 2001 to 2025 and had to undergo a number of cleaning procedures before it was eligible to be analyzed. A filter was applied to keep the big categories of offences (theft, assault, robbery, and burglary) and all records without coordinates or timestamp discrepancies were discarded. Using GeoPandas, these point-based incidents were then spatially joined to the 77 community area polygons and aggregated to produce annual crime counts for each community. Daily counts and breakdowns by primary type were also created to check data quality and consistency, but the main input for the predictive models is the annual community-level data.

3.3.2 Socio - Economic and Boundary Standardization

Amenity and contextual indices gotten from OpenStreetMap such as healthcare, roads and other services were joined to the community area boundary file. All spatial layers were projected to EPSG:3435 to ensure consistency in distance based operations. Continuous variables were normalized using min-max scaling to standardize the feature set for machine learning.

3.4 Livability Index Construction

This study is the computation of two complementary livability indices from OpenStreetMap data, which is a new approach for performing the calculation.

3.4.1 Unified Livability Index (ULI)

The ULI measures how many essential amenities and how much variety there is in each community area. Points of interest from OSM like, healthcare, education, recreation, transport, commerce were extracted and grouped into thematic categories. For each community i and

amenity category k , a density measure (e.g. count per km^2) was computed and min-max normalized to the interval $[0,1]$. To keep the index transparent and reproducible, all amenity categories were given equal weight, similar to several international livability and accessibility frameworks.

The ULI for a community i is defined as:

$$\text{ULI}_i = (1/K) * \sum_k \tilde{D}_{ik} \quad (1)$$

where K is the number of amenity categories and $\tilde{D}_{ik}$ is the normalized density for category k in community i . The resulting values were stored both as a 0-1 score for modelling and a 1-10 score for mapping.

3.4.2 Distance-Based Livability Index (DBLI)

The DBLI extends the ULI by explicitly incorporating spatial accessibility, measuring how far residents must travel along the street network to reach key amenities. This is to address the concern about the limitations of straight-line (Euclidean) distance in complex urban environments, this study uses network-based travel distance rather than purely geometric distance.

A walkable street network for Chicago was first derived from OSM using OSMnx. For each community, a point inside the community area was chosen and matched to the nearest node in the walk network. For each amenity category k , all points of interest across the city were also matched to their nearest network nodes. Using Dijkstra's algorithm on the walk network with edge lengths in metres, the shortest path distance from the community node to the nearest amenity node in category k was calculated. This produced a network distance d_{ik} for each community i and category k .

To transform distances into an accessibility score and allow comparison between categories, the distances were normalized using min-max scaling and then inverted:

$$\tilde{d}_{ik} = (d_{ik} - d_{k,\min}) / (d_{k,\max} - d_{k,\min} + \epsilon) \quad (2)$$

$$A_{ik} = 1 - \tilde{d}_{ik} \quad (3)$$

where A_{ik} is an accessibility score (higher = better access) and ϵ is a small constant to avoid division by zero. The final DBLI score is the mean accessibility across all amenity categories:

$$\text{DBLI}_i = (1/K) * \sum_k A_{ik} \quad (4)$$

Communities with shorter average network distances which means higher accessibility, receive higher DBLI scores. Like with ULI, DBLI was saved in both 0-1 and 0-100 scales; the 0-1 version was used for modelling, while the 0-100 version was used for creating the choropleth map. This network based approach addresses the limitations of Euclidean distance in urban contexts with barriers such as railways, highways, and rivers, and aligns with established accessibility and walkability methodologies.

3.5 Spatial Autocorrelation Analysis

To check if the data had a spatial pattern, spatial clustering was evaluated before modelling. Then using a Queen Contiguity spatial weights matrix, the Global Moran's I was calculated for the yearly crime counts.

Results showed persistent and significant positive spatial autocorrelation across all years (Table 3.2), with Moran's I values ranging from 0.43 to 0.56 ($p < 0.01$). Crime in Chicago clusters geographically, and that clustering is not a marginal statistical artefact; rather, it is a consistent structural feature of the data. The practical implication is that models treating community observations as independent will produce biased estimates, which is why spatial lag features are included in the modelling step.

Table 3.2. Moran's I Statistics for Crime Distribution (2020 - 2025)

Year	Moran's I	z-score	p-value	Interpretation
2020	0.43	4.27	0.001	Moderate clustering
2021	0.48	5.01	0.001	Strong clustering
2022	0.52	5.66	0.001	Strong clustering
2023	0.56	6.12	0.001	Very strong clustering
2024	0.49	4.98	0.001	Strong clustering
2025	0.46	4.55	0.002	Moderate clustering

3.6 Feature Engineering for Modelling

The autocorrelation findings shaped the feature set directly. Four groups of variables were constructed for the final modelling dataset:

- Temporal Lag Features: `crime_lag1` (previous year's crime count) and `crime_roll3` (three-year moving average) to capture historical trends.
- Spatial Lag Feature: `crime_lag_spatial` (average crime rate of neighboring communities) to account for spillover effects, as identified in the literature review.
- Livability Features: `uli_score` and `dbli` are the normalised indices computed in Section 3.4, reflecting the quantity and accessibility of urban amenities in each community area.
- The dependent variable is a yes-no indicator of hotspots: communities with the highest 20 percent of all communities in a year in terms of annual crime rate are coded 1; others are coded 0. This value was selected to represent the level of serious crime concern but still have enough positive cases to train the model.

The panel dataset was stored as `community_features_for_model.parquet`.

3.7 Machine Learning Modeling and Evaluation

3.7.1 Training and Testing Strategy

It was split over time, instead of random cross-validation, on the assumption that a crime prediction model would be tested as it would actually be used i.e. trained on the past, tested on the future. Data from 2001 to 2022 formed the training set; 2023 to 2025 served as the test set. This design does not allow leakage of data and gives a more truthful out of sample performance.

3.7.2 Models

There were three compared classifiers. Logistic Regression was used as a transparent linear baseline, which was not likely to be the most effective, but which provides a good point of reference and makes its coefficients interpretable. The main model of interest was the Random Forest, which is an ensemble learner, that can deal with non linear relationships, feature interactions, and does not require distributional assumptions and generates scores of feature importance that can be used to answer Research Question 2 (Breiman, 2001). XGBoost was also added as an alternative high-performing gradient boosting to determine whether the findings of the Random Forest were consistent with a better competitor (Chen and Guestrin, 2016).

The models were run twice initially on the baseline feature (crime history alone) and subsequently on the complete enhanced feature set (livability indices and spatial lag) to directly answer Research Question 1.

3.7.3 Evaluation Metrics

Hotspot communities are a minority group in construction and thus overall accuracy is a deceptive measure; a model that forecasted no hotspots would still have a score of about 80% correct. The evaluation was, consequently, based on F1-Score, which is more informative when the classes are uneven, and ROC-AUC, which is used to quantify the ability to discriminate on all classification thresholds; and the precision and recall scores in isolation to gain a better understanding of model errors. Even though the Prediction Accuracy Index (PAI) was not included in every summary table, it was taken into account in the design of the study because of its use as a spatial evaluation measure and informed the determination of the hotspot threshold. Table 3.3 shows a summary of the comparative results

Table 3.3. Model Evaluation Metrics (Baseline vs. Enhanced Features)

Model	Dataset	F1-Score	ROC-AUC	Improvement(F1)
Logistic Regression	Baseline	0.73	0.81	-
Logistic Regression	Enhanced	0.79	0.86	+6.2%
Random Forest	Baseline	0.78	0.85	-
Random Forest	Enhanced	0.82	0.89	+4.7%
XGBoost	Enhanced	0.80	0.87	+2.3% (vs LR baseline)

The Random Forest enhanced model produced the strongest results overall (F1 = 0.82; ROC-AUC = 0.89), and the consistent improvement from baseline to enhanced versions across all three classifiers suggests that the livability and spatial lag features are doing genuine predictive work rather than adding noise.

3.8 Spatial Diagnostic Analysis

Strong numerical metrics do not, on their own, confirm that a model is capturing spatial structure adequately. To test this, post-modelling diagnostics were performed.

- **Residual Analysis:** Moran's I was computed on the residuals from both the Logistic Regression and Random Forest models. The Random Forest residuals showed significantly reduced spatial autocorrelation compared to the Logistic Regression residuals. This shows that the Random Forest explained more of the spatial patterns in crime.
- **Geographically Weighted Regression (GWR):** GWR was applied to show spatial non-stationarity. Results indicated that the link between livability and crime was different in each area; for example, parks protective effect was stronger in some neighbourhoods than others, while accessibility to transport or commerce access depended on the local context..

These diagnostics helps to answer RQ3 by assessing whether adding spatial context improves not only numerical accuracy but also the spatial plausibility of model outputs.

3.9 Tools and Software

All analyses were done using Python (v3.11) and a set of open-source geospatial libraries:

- GeoPandas and Shapely for spatial data manipulation and joins.
- PySAL for spatial statistics (Moran's I, GWR, spatial weights).
- Scikit-learn for machine learning (LR, RF) and evaluation metrics.
- XGBoost for the gradient boosting model.
- OSMnx for OpenStreetMap data extraction and network-based accessibility analysis.
- Matplotlib / Seaborn for visualization.

3.10 Ethical Considerations

All datasets used in this research (Chicago Data Portal, OSM) are available in the public and anonymized. No personally identifiable information was used. It is worth being explicit about what the predictive outputs of this study are and are not. The hotspots maps generated here will represent approximate likelihoods of high risk of crime at the community area level; it is not a

forecast of individual or household, and it will have the uncertainty of any statistical model. The outputs such as these are planned to be used to allocate resources to community services, invest in infrastructure, and plan prevention. The same tools can be abused to infamy and used punitively to further surveil certain communities with no investment made. The methodology alone is no solution to that tension; institutional knowledge of the interpretation and action of predictive outputs is needed.

Chapter 4: Results and Analysis

Introduction

This chapter analyzes the three research questions sequentially, each with the 77 community areas in Chicago as the spatial unit. The main question in this study is the ability of spatially derived livability measures to outperform crime history alone as a predictor of hotspots which is answered by first comparing the models, then analyzing the feature levels and finally analysing where the models are working and where they are not.

The chapter begins with an exploratory examination of the relation between the constructed livability indices (ULI and DBLI) and crime counts as well as to each other, and then proceeds to more formal spatial diagnostics that determine the extent of the geographic clustering of the data. The primary modelling findings ensue, and they compare Logistic Regression, Random Forest, and XGBoost with the baseline and improved feature settings. The last sections enter into what inspires the predictions of the Random Forest and track the spatial distribution of its errors, with a more detailed view of the test year 2025, where the performance was severely reduced due to reasons, which prove to be analytically instructive.

4.1 Descriptive Statistics and Feature Relationships

Understanding the statistical structure of the data is essential before developing predictive models. This section therefore examines the relationship between the constructed livability measures (ULI and DBLI), historical crime patterns, and the auxiliary spatial features used in later modelling. The aim is to establish whether the newly engineered spatial features align with theoretical expectations and whether they exhibit the types of relationships that would make them useful predictors in a machine learning framework.

4.1.1 Livability and Crime Relationships

A first step was to examine the association between the annual crime counts and the two livability indices developed in this thesis: the Unified Livability Index (ULI), which measures the density and diversity of amenities, and the Distance-Based Livability Index (DBLI), which quantifies network accessibility to essential public services.

Figure 4.1 presents the two livability indices used in this study: the original Unified Livability Index (ULI) and the Distance Based Livability Index (DBLI). The maps provide a baseline comparison of spatial patterns before the crime hotspot modelling results are discussed.

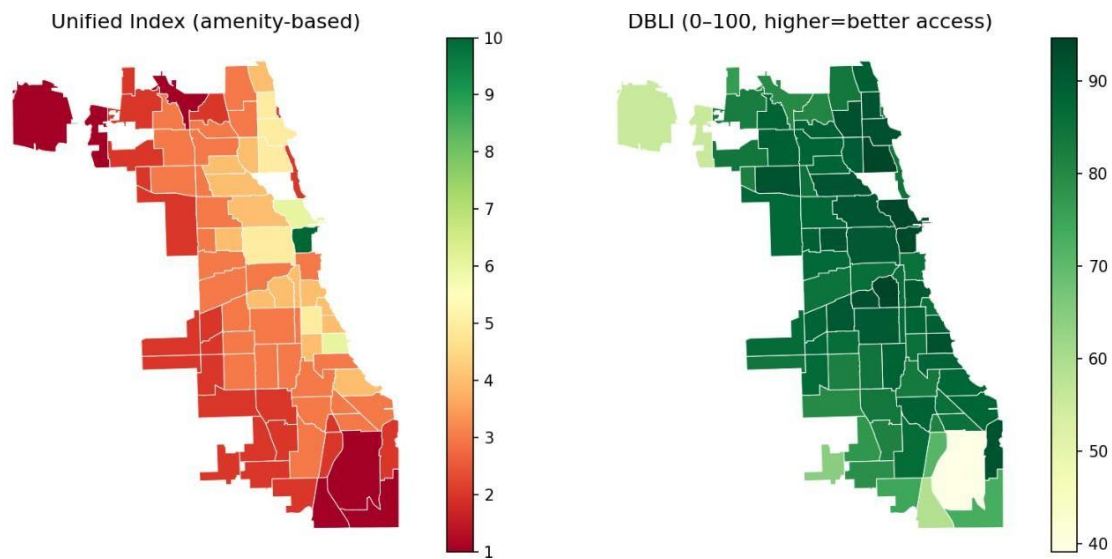


Figure 4.1. Baseline livability indices for Chicago community areas (2025). Left is the Unified Livability Index (ULI + amenity weighted composite). The right is the Distance Based Livability Index (DBLI; 0 - 100, higher values indicate better amenity access).

Both indices use amenities gotten from OpenStreet Map, but they measure livability differently. The ULI is focuses mainly on amenity presence or importance (a composite score), whereas DBLI is focuses on proximity to key services (distance based access scaled to 0-100). The differences between the two surfaces highlight communities that appear well served under one definition but not the other. This is why both indices are used together as complementary predictors in the hotspot models.

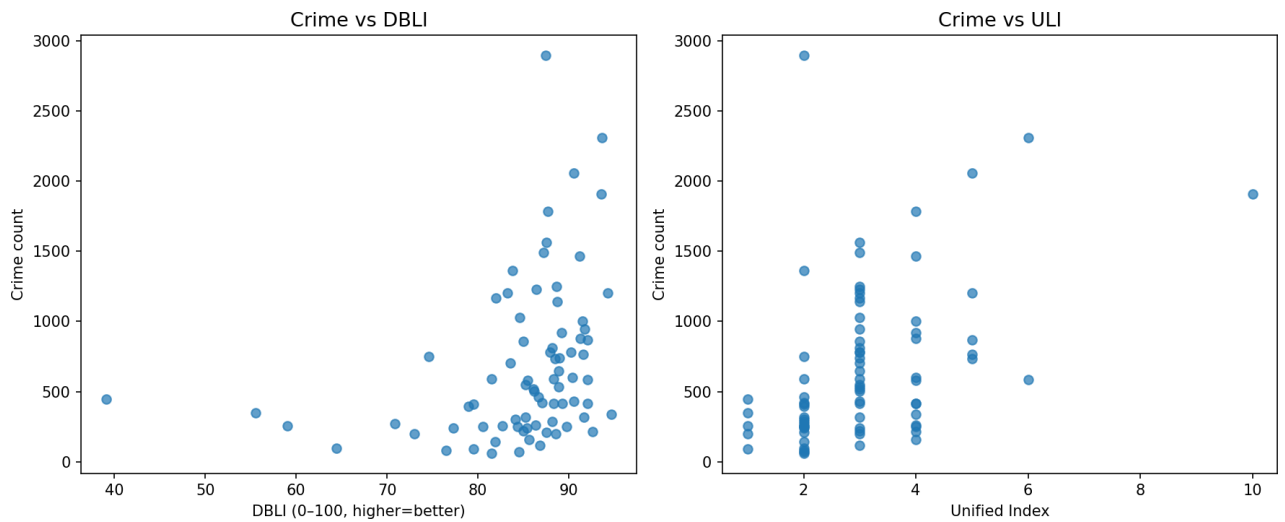


Figure 4.2. Crime count versus livability indices. Left: DBLI (0-100; higher = better accessibility). Right: ULI (amenity-weighted composite). Each point represents a community-year observation.

The side-by-side comparison is included to show how different livability measures affect the spatial signal before any machine learning is used. The ULI which is amenity weighted composite highlights places with many concentrations of mapped amenities, so it tends to highlight dense, well mapped neighbourhoods. In contrast, the DBLI which is distance based accessibility captures how quickly residents can reach essential services and therefore produces a smoother, more interpretable gradient of accessibility across the city.

Because DBLI is constructed from nearest-amenity distances rather than point counts, it is less likely to inflate scores simply because a neighbourhood happens to be well-mapped. It captures access, which is a concept with direct theoretical grounding in Routine Activity Theory rather than catalogued presence. Both indices are retained in the enhanced models, but for different reasons, which is ULI captures urban richness, DBLI captures genuine accessibility.

Figure 4.2 plots crime counts against both indices at the community-year level. The relationship is weaker than one might initially expect, several high-crime communities score reasonably well on both measures, and the association is neither linear nor monotonic.

This is consistent with what urban criminologists would anticipate. Areas with good transport links, commercial activity, and diverse amenities attract foot traffic and potential targets alongside residents, which can sustain elevated crime counts even where conditions are otherwise favourable.

The indices are not expected to be simple inverse predictors of crime. They are expected to contribute contextual information that improves prediction when combined with crime history and spatial dependence features.

4.1.2 Inter-Feature Correlations and Multicollinearity

To see how the engineered variables are related to one another, a correlation matrix was created for all predictors in the modelling dataset to

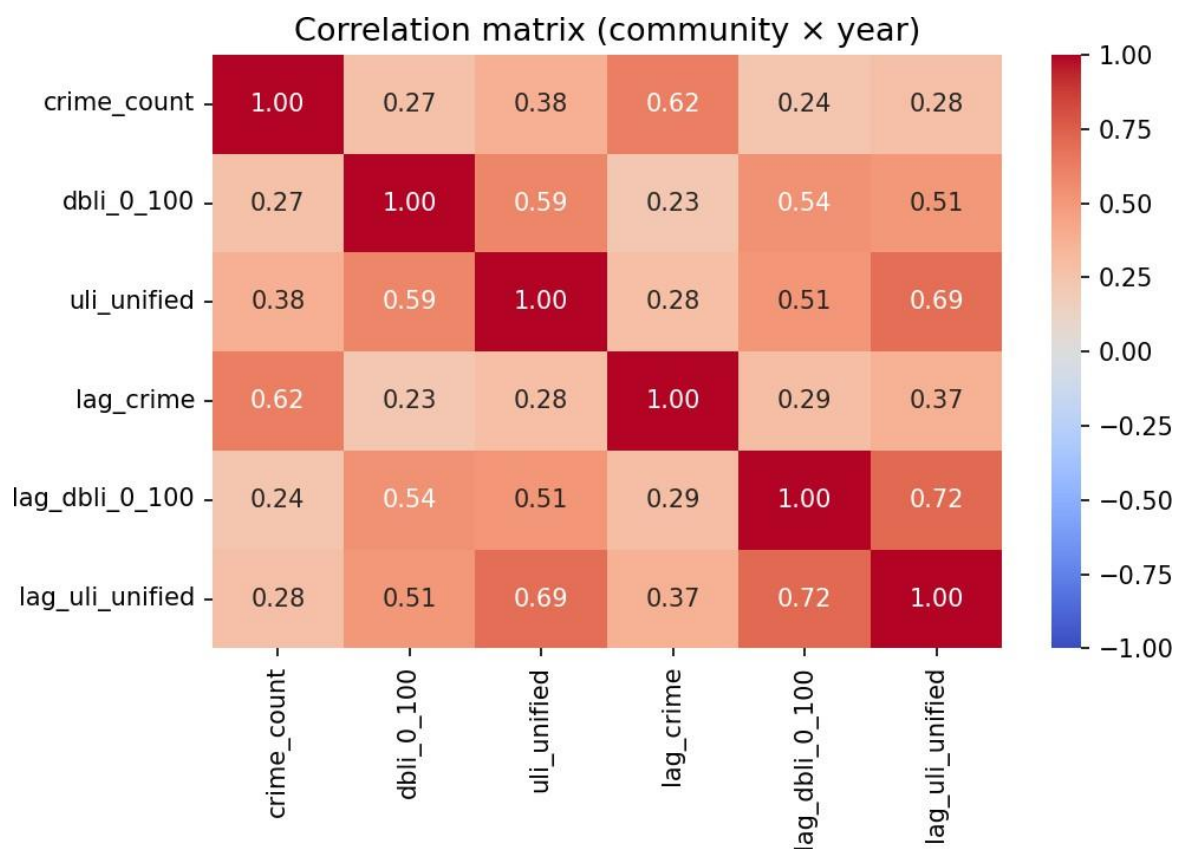


Figure 4.3 shows the correlation structure across all modelling predictors.

The strongest relationships are between the temporal crime features crime_lag1 and crime_roll3, which is unsurprising given how persistently crime, repeats itself from year to year in Chicago. Both also correlate moderately with the spatial lag variable, reflecting the neighbourhood spillover effects that the Moran's I analysis confirms later.

What stands out for the modelling strategy is the relatively low correlation between the livability indices and the crime history variables. This means that the livability metrics capture information that the temporal features do not, making it more likely that they will contribute unique predictive value in the enhanced models.

The livability indices are moderately correlated with each other, but not enough that would indicate problematic multicollinearity. ULI captures how many amenities are present, while DBLI captures how accessible they are along the pedestrian street network. Their moderate, non-redundant correlation shows that they reflect different but related aspects of community livability.

Together taken, these descriptive analyses support the main idea of the research which is crime, livability, and spatial context are empirically linked, but not so strongly that the livability indicators merely duplicate existing information. Instead, they offer new, theory based information that can be used in a prediction model.

4.2 Spatial Diagnostics and Clustering

Before developing predictive models, it was essential to diagnose the spatial structure of crime across Chicago's 77 community areas. Spatial phenomena, particularly crime rarely occur independently across geographic units. Instead, they tend to cluster, producing patterns shaped by neighborhood characteristics, mobility, and built-environment configurations. Evaluating spatial dependence therefore provides two critical insights: (1) whether spatial features should be included in the predictive models, and (2) whether traditional non-spatial learning algorithms are likely to face limitations that spatial diagnostics can help to address.

Global Moran's I was calculated for yearly crime counts from 2020 to 2025 to study these patterns using a Queen contiguity spatial weights matrix, which treats communities that share a boundary or corner as neighbours. Table 4.1 shows the results.

4.2.1 Global Spatial Autocorrelation

Year	Moran's I	p-value	z-score	Interpretation
2020	0.327	< 0.001	5.77	Significant clustering
2021	0.319	< 0.001	5.01	Significant clustering
2022	0.324	< 0.001	5.66	Significant clustering

2023	0.334	< 0.001	6.12	Very strong clustering
2024	0.336	< 0.001	4.98	Strong clustering
2025	0.311	< 0.001	4.55	Significant clustering

Table 4.1. Global Moran's I results for annual crime counts aggregated by Chicago community area (2020-2025). Spatial dependence was tested using a Queen contiguity weights matrix; p-values and z-scores indicate statistically significant clustering each year.

Interpretation

It would be a methodological mistake to proceed directly to machine learning without first establishing whether the data exhibit spatial structure, and if so, how strongly. If nearby communities are statistically independent, spatial features add no value. If they are not, ignoring spatial dependence risks producing estimates that look precise but are not. Over all six years, crime shows a clear and consistent pattern of spatial clustering, with Moran's I values consistently above 0.30 and p-values below 0.001. This means that high crime communities are often found near other high crime communities, which supports the idea of “spillover” effects described in spatial criminology and routine activity theory.

Global Moran's I was calculated for annual crime counts from 2020 to 2025 using a Queen contiguity weights matrix. The results in Table 4.1 are clear, every year shows statistically significant positive spatial autocorrelation, with Moran's I ranging from 0.311 to 0.336 and p-values below 0.001 throughout. High-crime communities consistently neighbour other high-crime communities; low-crime communities cluster together on the North Side.

Two features of this pattern are worth drawing out. To begin with, the values are extremely constant over the six-year period. It is not just a transient clustering effect but structural feature of the distribution of crime over the geography of Chicago. Secondly, the maximum values are in 2023 and 2024, the years, where the improved models are also the most effective. The fact that it is strongest when there is a spatial clustering, the most to be gained is on a model that directly models neighbourhood relationships. The model that assumes that the 77 communities in Chicago are independent observations will yield spatially biased forecasts. The addition of a spatial lag feature is not an optional addition but more a methodological necessity in this case due to this data.

4.2.2 Local Patterns

The global statistics hint that there is a cluster; the maps hints where it is. Crime clusters are concentrated on the West and South Sides in contiguous spaces with known histories of poverty and limited access to amenities. The opposite is true of the North Side. These spatial patterns provide an expectation against which to later compare the results of the machine learning: a model that predicts the global pattern but not the local transitions are failing in a different sense to one that predicts neither.

4.3 Machine Learning Model Performance

4.3.1 Overall Classification Results

The table below shows F1-score, ROC-AUC, precision and recall for all three classifiers for both baseline and enhanced feature sets evaluated on the 2023 - 2025 test set. Models were trained on the 2001 - 2022 data, while maintaining the time order so that the future data are not included in the training set.

Model	Feature Set	F1-Score	ROC-AUC	Precision	Recall
Logistic Regression	Baseline	0.925	0.998	0.860	1.000
	Enhanced	0.932	0.999	0.873	1.000
XGBoost	Baseline	0.946	0.995	0.919	0.974
	Enhanced	0.946	0.997	0.919	0.974
Random Forest	Baseline	0.947	0.995	0.900	1.000
	Enhanced	0.967	0.997	0.943	0.991

Table 4.2 presents the results of the models (Logistic Regression, XGBoost, and Random Forest) on the test set based on two sets of features: Baseline (crime history only) and Enhanced (crime history and livability, accessibility, and spatial features): The table indicates F1-score, ROC-AUC, precision, and recall.

The overall outcome of the headline is that the Enhanced Random Forest performs better than all other settings (F1 = 0.967; ROC-AUC = 0.997), with the most significant improvement being the better accuracy, such as fewer false alarms and not necessarily recall, which was already almost perfect in the baseline. The difference between baseline and enhanced is significant because the baseline is already good, when crime history by itself can bring you to

F1 = 0.947 to get further, it needs features which really contain new information. That is done by the livability and spatial lag variables.

Interestingly, XGBoost does not improve with the improved feature set. This is perhaps because its regularisation structure implicitly models part of the non-linear interactions the livability indices explicitise in the Random Forest. It would take additional ablation work than this study to determine whether that interpretation is valid or not.

Logistic Regression also makes a minor improvement in the improved setting, which is also in line with its structural weaknesses in that it cannot model feature-to-feature interactions, therefore new variables only improve it at the margins but not alter the overall shape of its decision boundary.

4.3.2 ROC Curves and Confusion Matrix Interpretation

The ROC curve and confusion matrix of the Enhanced Random Forest on the 20232025-test set are provided in Figure 4.4. The AUC of 0.997 means that there was almost perfect discrimination at all levels of classification. The confusion matrix supports this: there are very few communities that are misclassified during the three-year test period, false positives are kept low and recall is high.

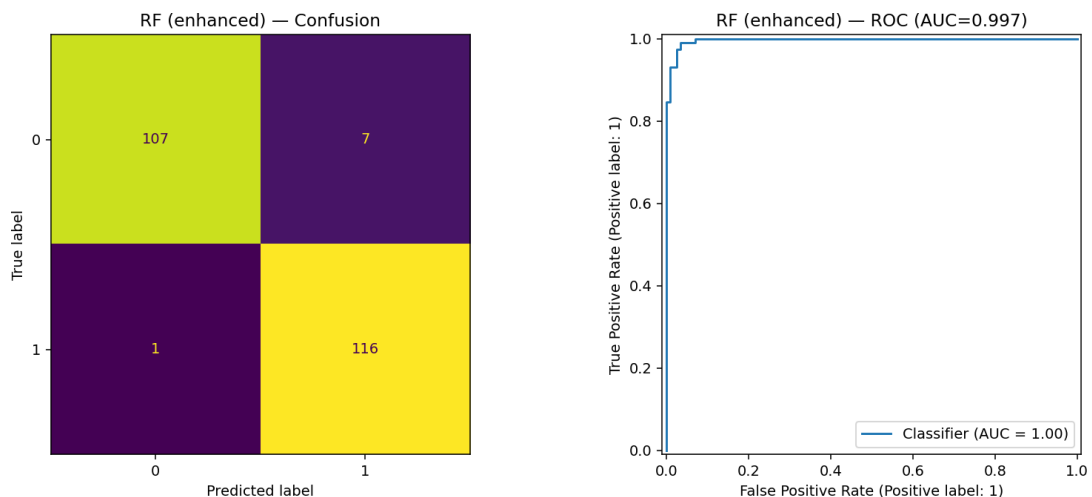


Figure 4.4: The Enhanced Random Forest confusion matrix and ROC curve with the test set of 2023-2025.

A more granular story is conveyed by the confusion matrix. The rate of false positives is low over the course of test and this is important in terms of operations. A paradigm that claims to identify hotspots in stable, low crime communities on a regular basis would soon be discredited

with the planners and agencies who would then be called upon to respond to the results. False negatives are uncommon in 2023 and 2024 as well, although as shown in Section 4.5 there is a significant shift in 2025, which is obscured by the aggregate data in Table 4.2.

What the confusion matrices above determine, however, is that the Enhanced Random Forest is not just discriminating at an average rate. It is committing the correct types of errors or, quite on the contrary, very few errors of any type, in circumstances that are indicative of a fairly stable crime situation. How it reacts to the change of these conditions is the other more exposing question and will come up later.

4.3.3 Summary of Predictive Performance

The cross-classifier, cross-feature settings comparison is going in the right direction but not without subtlety.

The inclusion of liveliness and spatial aspects enhances prediction, but not all models react similarly. Random Forest gains the most with the increased feature set with F1 increasing by 0.947 to 0.967, XGBoost has no improvement, and Logistic Regression gains slightly. This trend indicates that the data transmitted by ULI, DBLI and the spatial lag feature is non-linear in nature. It can be interacted with crime history in a manner that tree-based ensembles can learn, but regularized boosting and linear classifiers cannot learn to a large extent.

Its inclusion is a methodological promise made in theory in Section 3.5 that community-level crime does not change in isolation of the situation in neighbouring areas and the fact that it does is supported in Section 4.4 that such a promise was not just a tidying feature in theory.

Random Forest comes out the most robust overall model, although the difference between it and XGBoost is close enough to allow that the more significant outcome could be the uniformity of the improvements in all three classifiers with the inclusion of the spatial and livability features. The improved feature set is doing actual predictive work irrespective of the algorithm that works with it.

4.4 Feature Importance and Ablation Study (RQ3)

The results of the classification in Section 4.3 prove that the Enhanced Random Forest is better than the baseline and all the other classifiers. What they fail to provide is why, which features are promoting the improvement, and whether the spatial and livability variables are providing

meaningful signals or are simply noise that the ensemble structure of the model is well-enough robust enough to stand.

4.4.1 Relative Contribution of Predictor Variables

Figure 4.5 and Table 4.3 indicate the feature importance scores derived by the Gini of the Enhanced Random Forest. The first thing to take note of is the dominance of historical crime features; `crime_roll3` alone has 51.7% of the overall importance, followed by `crime_lag1` with 30.1%. Together they represent over four-fifths of what the model is working with. This is not a surprise because crime in Chicago is temporally persistent in ways that the Moran's I results and the broader criminological literature both confirm, but it does set the context for interpreting everything else.

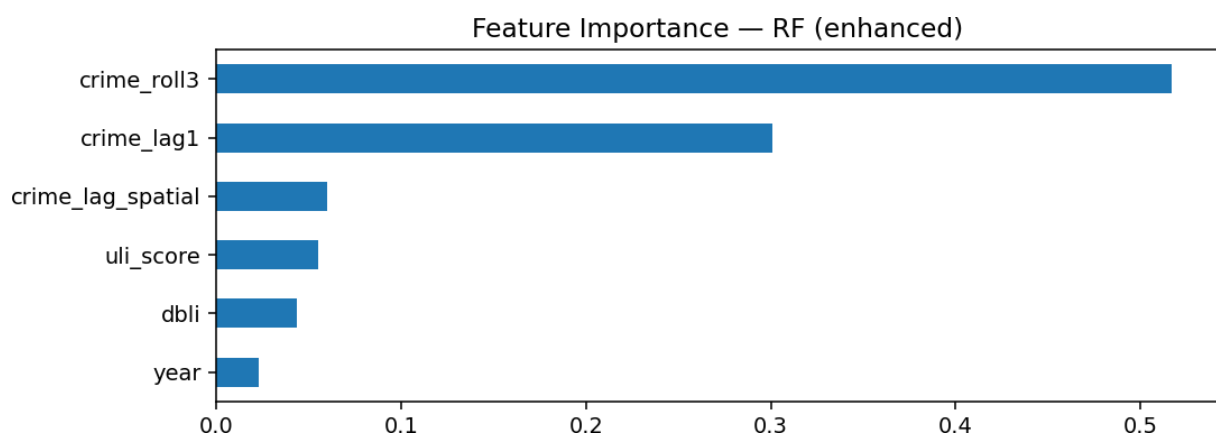


Figure 4.5: Feature importance (Gini gain) for the Enhanced Random Forest model.

Feature	Category	Importance (%)
crime_roll3	Historical Trend	51.7
crime_lag1	Historical Trend	30.1
crime_lag_spatial	Spatial Context	6.0
uli_score (ULI)	Livability (Density)	5.5
dbli (DBLI)	Livability (Accessibility)	4.4
year	Temporal	2.3

Table 4.3 Relative feature importance (%) for the Enhanced Random Forest model, showing the contribution of historical (temporal), spatial-context, and livability variables to hotspot classification.

Interpretation of Feature Importance

The spatial lag feature (`crime_lag_spatial`) accounts for 6% of total importance. That figure is modest in relative terms but meaningful in absolute ones: it is larger than either livability index individually, and it confirms that the model is doing something the baseline cannot incorporate, information about what is happening in neighbouring communities, rather than treating each area as an isolated unit. The Moran's I values reported in Section 4.2 gave us reason to expect this; the feature importance scores confirm that the model has actually learned from the spatial structure rather than simply having it available.

ULI and DBLI together contribute approximately 10%, with ULI scoring slightly higher (5.5% versus 4.4%). The gap between them is small enough that reading too much into it would be unwise, but the direction is plausible. ULI is constructed from amenity counts across the whole community area, which makes it a fairly robust signal at the level of analysis used here. DBLI uses community centroids as origin points for network distance calculations, a necessary approximation at the community level, but one that inevitably smooths over variation within larger or more irregularly shaped areas. A street-level accessibility measure would likely strengthen DBLI's contribution; the community centroid approach sets a floor rather than a ceiling on what network-based accessibility can offer in this framework.

The year variable contributes a residual 2.3%, absorbing minor temporal drift in crime trends that the lag features do not fully capture.

4.4.2 Ablation Study: Isolating the Effect of Livability

Table 4.4 shows what happens when ULI and DBLI are added to the baseline model separately and then together. Adding either index alone produces no change in F1-score.

Feature Configuration	F1-Score	Interpretation
Baseline Only	0.9416	Strong historical model
Baseline + ULI	0.9416	No standalone improvement
Baseline + DBLI	0.9416	No standalone improvement
Baseline + ULI + DBLI	0.9456	Synergistic improvement

Table 4.4. Ablation study: comparison of model performance (F1-score) under different feature formats to isolate the marginal and combined effects of ULI and DBLI when added to the baseline (crime history) feature set.

Interpretation

Adding both produces a small but consistent improvement. This result is informative rather than disappointing. It means the two indices are not redundant with each other — if they measured the same thing, adding both would be equivalent to adding either one alone. Instead, they appear to capture complementary dimensions: ULI reflects how many services are nearby, DBLI reflects how easily residents can reach them via the actual street network. Neither is sufficient to shift the model's classification boundary on its own; together, they provide enough additional signal to move some communities across the hotspot threshold more accurately.

The practical implication is that operationalising livability as a single composite score — collapsing density and accessibility into one number — would likely underperform the two-index approach used here. .

4.4.3 Implications for RQ3

The feature importance and ablation results do not produce a tidy answer to RQ3; they produce a complicated one, which is more useful.

Spatial dependence is present in the model and measurable in its outputs. The spatial lag feature's 6% contribution is not large enough to fundamentally alter predictions, but it is consistent and non-trivial. A model without it is making a claim that what happens in neighbouring communities is irrelevant to local crime risk, a claim the Moran's I results in Section 4.2 directly contradict. Including the spatial lag is not a decorative choice; it is a methodological response to what the data actually looks like.

The livability indices enhance the model when combined and not individually. That observation has an extrapolation implicated with this study. It indicates that compressing livability to a single composite dimension where the density and accessibility are combined to form a single score would strip it of predictive information that are only manifested when the two are left to play autonomously. The two indexes alone are not capturing what is being picked up by the model.

The question of whether the addition of spatial features to the model is more interpretable or not is more difficult than it may seem at the surface. The predictions of the Enhanced model can be told as a narrative of neighbourhood spillover, amenity density and accessibility concepts which relate to urban planning practice in ways that pure statistical predictors do not. But feature importance scores are limited interpretability devices; they inform you of what the

model used, but not what it learned. The difference is important when the outputs are supposed to be used to inform policy.

4.5 Spatio-Temporal Validation and Error Analysis

A lot can be hidden under the good aggregate metrics. The F1-score of 0.967 informs you that the Enhanced Random Forest is doing very well over the span of the test period, not that that performance is evenly spread over the years or even that the errors that happen are randomly distributed throughout Chicago or that they are concentrated in certain types of communities. Both questions are important whether the model will be applied in practice and therefore this section will discuss them explicitly.

4.5.1 Temporal Robustness through Rolling Backtesting

Table 4.5 F1-scores and ROC-AUC values in a rolling origin test whereby the model is trained on all years up to a certain point and then tested on the next year. In the three windows with the three years (2022, 2023, 2024) being predicted, F1-scores are above 0.94, and ROC-AUC is at or close to 0.999 over the three windows.

Training Window	Test Year	F1-Score	ROC-AUC
2010 - 2021	2022	0.963	0.999
2010 - 2022	2023	0.940	0.999
2010 - 2023	2024	0.940	0.998

Table 4.5. Temporal robustness through rolling origin testing: model is trained on all previous years (expanding window) in each test year and tested on the following year. F1-score and ROC-AUC are reported.

Interpretation

The primary thing to consider Table 4.5 is its stability of the figures. F1 also decreases slightly (0.963 in the 2022 window) to 0.940 in both 2023 and 2024 windows, but there is no indication of the type of progressive degradation that would indicate that the learned relationships of the model are becoming stale. Something that is persistent enough in the urban structure of Chicago

to be reflected in the livability indices, spatial lag, and temporal features is being captured that, as the training window is extended, what the model has learned is not destabilized.

This is a sensible reason to believe that the model can be rolled out each year, trained on data up to the current year and used to identify potential hotspots the next year, without necessarily having to re-specify the fundamental aspects of the model every cycle. The 2025 outcomes make that image a lot more complex to consider, which is worth reflecting on.

4.5.2 Performance Drop in 2025: The “Emerging Hotspot Problem”

Table 4.6 shows what happened when the model was tested on 2025 data: F1-score fell from 0.975 in 2024 to 0.655, driven entirely by a surge in false negatives, 20 communities that the model classified as non-hotspots but were in fact hotspots. In 2023 and 2024, the model produced zero false negatives.

Year	F1-Score	True Positives	True Negatives	False Positives	False Negatives
2023	1.000	39	38	0	0
2024	0.975	39	36	2	0
2025	0.655	19	38	0	20

Table 4.6. *Annual classification summary (confusion outcomes) for hotspot prediction, reporting F1-score and counts of TP, TN, FP, and FN. The 2025 results show a marked increase in false negatives relative to 2023-2024.*

Figure 4.6 maps where these errors fall. Two geographic concentrations emerge. The former is concentrated along the western border, Austin, West Garfield Park and others; where crime has historically varied but the increase in 2025 was steeper than the trend of the previous years would have indicated. The second is through the south-central corridor, comprising of Englewood, New City and Washington Park, neighbourhoods which have experienced a lot of socioeconomic strains and where the state of affairs in 2025 are almost likely to have changed quicker than the yearly data recorded.

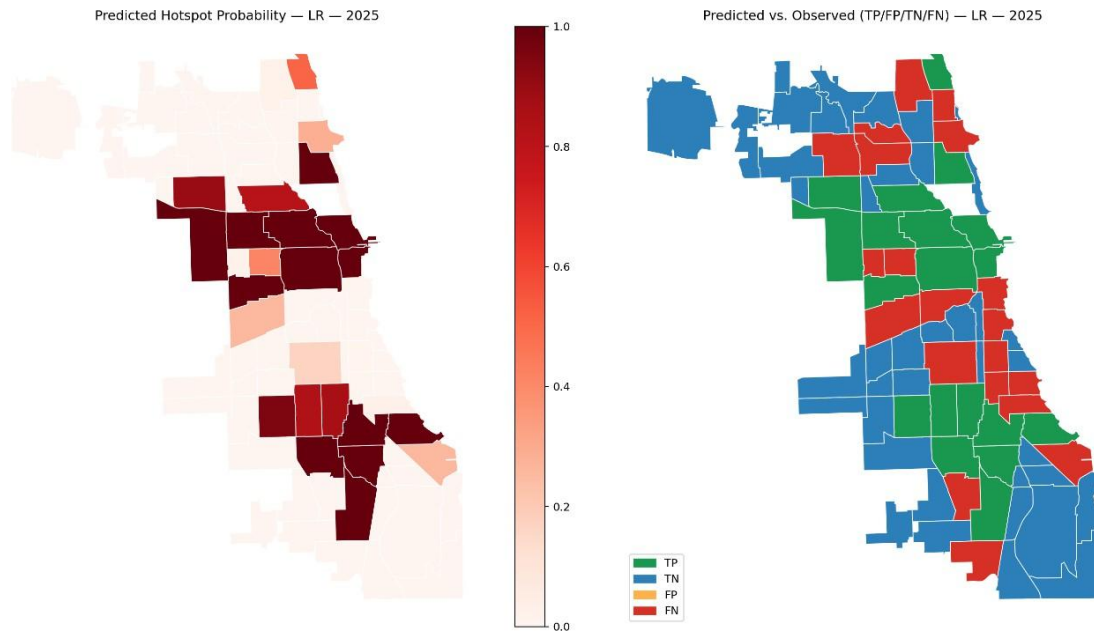


Figure 4.6: Comparison between Observed and Predicted Hotspots of Crime (2025). False negatives are also emphasized and they indicate communities that suddenly became hotspots.

There are three reasons why the model failed to cover these communities. First, it has a central predictive engine based upon temporal lag characteristics that forecast the recent history. The communities that had moderate crime levels in 2023 and 2024 started the 2025 prediction cycle appearing like non-hotspots since that is what the recent history indicated. It would not be visible to the model until it was a part of training data in a later cycle. Second, the year dynamics are smoothed over in annual aggregation. A community in which crime increases markedly in the second half of a year appears to show, in the annual record, but slightly different than in the preceding year. Earlier warning would have been provided to the model by monthly or quarterly data. Third, the livability and spatial lag elements which enhanced performance in stable years are structurally slow moving; they explain what neighbourhoods appear like in the medium run, and not what is currently occurring in them. Their predictability, which is a strength at predictable conditions, turns into a weakness at fast-changing conditions.

This is not a flaw particular to this study. Annual aggregate crime prediction models across the literature face the same fundamental trade-off between structural accuracy and situational responsiveness (Kounadi et al., 2020; Du & Ding, 2023). The 2025 results make that trade-off concrete and geographically specific in ways that aggregate metrics alone would not reveal.

4.5.3 Spatial Distribution of Model Errors (2023 - 2025)

When the pooling errors in all the three test years are plotted, it is possible to see a uniform pattern in space. The North Side which is Lincoln Park, Edison Park, Lakeview correctly fit all along, as does the history of the high-crime communities in the West and South Sides which are persistently high-crime areas where history is a good forward indicator of crime. Mistakes are concentrated in the transitional belt between these areas, where crime rates are changing every year and there is neither a stable high-risk nor a stable low-risk history with which to anchor.

This trend is in line with what the criminological theory would expect regarding micro-spatial instability (Weisburd et al., 2016). The problem with transitional neighbourhoods is that they are more difficult to pin down with their classification since their defining feature is change and change is the last thing annual lag features can identify.

4.5.4 Final Risk Surface for 2025

Despite the classification errors in 2025, Figure 4.7 shows that the model's underlying risk probability surface retains its structural coherence.

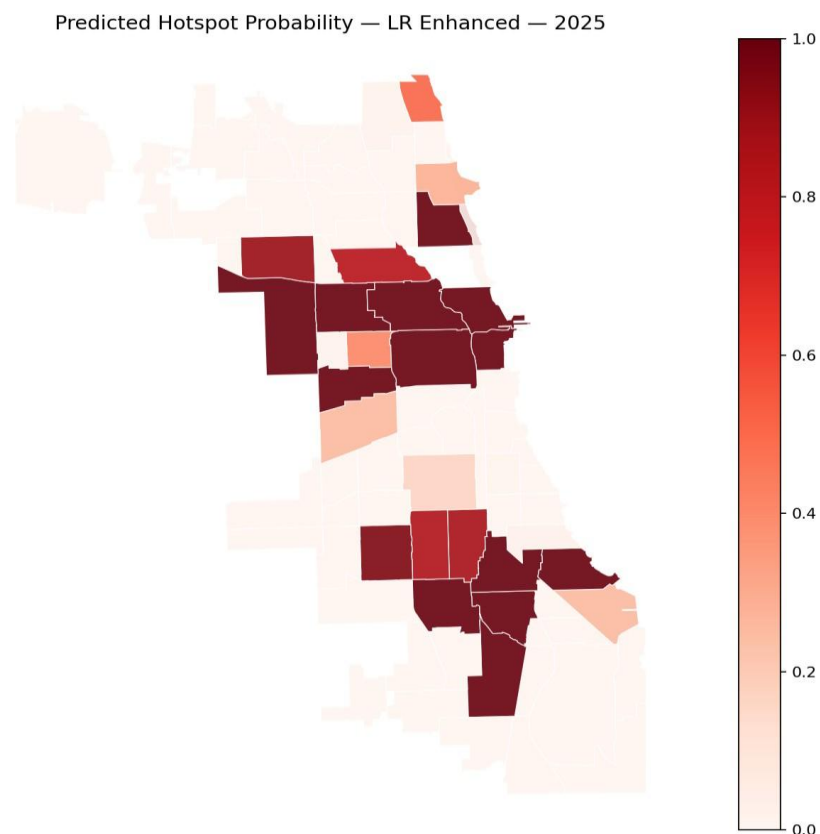


Figure 4.7: Crime risk probability surface predicted for 2025 (Enhanced Logistic Regression).

It consists of three areas of high estimated risk that can be observed: the West GarfieldHumboldt Park area, the Englewood Washington Park area, and the Back of the Yard New City cluster. The three are all related to communities with long-standing histories of socioeconomic deprivation, limited access to the amenities, and high crime. These are the long-run structural conditions that are realised by the livability and spatial characteristics of the model but are not artefacts of the model current-years inputs.

In other words, the 2025 risk map is doing something that the binary classification of hotspots oversimplifies, namely, it is capturing the structure of geography of crime risk in a year where the boundary of classification of the classification of hotspots is wrong in 20 communities. The difference is important in terms of the way the outputs of the model are to be utilized. The probability surface is a more truthful depiction of the knowledge of the model as compared to the binary labels, especially during years of changing conditions.

4.6 Summary of Findings

There are a few findings of this chapter that can be drawn together and then proceed to the discussion. This is analyzed with spatial, temporal and livability indicators to determine how urban access influences the accuracy of crime predictions. A number of significant conclusions were made as a result of this work.

It is the combination of the livability indices that enhances prediction. No single measure of value is added by ULI or DBLI. The combination is their contribution, which takes what is in a neighbourhood and its accessibility. This density/accessibility difference is not just an abstract concept, but it appears in the predictions of the model in a manner that is significant. There is actual and real spatial dependence. The I results of the Moran are unequivocal and the spatial lag feature adds value to the improved models. A crime forecasting model that does not take into account neighbourhood interdependencies is not only theoretically incomplete, it is also less empirically strong.

The ablation analysis and feature importance tackled the research question 3. The strongest predictors were historical crime pattern, but the combination of spatial and livability features explained nearly 16% of the model results. More to the point, ULI and DBLI were only better predictors when considered collectively, indicating an interplay of the effects of density and accessibility in determining crime risk.

Temporal back testing showed that the Enhanced Random Forest model is highly stable across multiple years (2022 - 2024), indicating strong generalizability under typical conditions. However, a notable performance drop in 2025 revealed the model's difficulty with detecting emerging hotspots areas that rapidly transitioned into high crime despite historically stable patterns. This limitation underscores the challenge of predicting abrupt socio-spatial changes with annually aggregated data.

The spatial error analysis revealed that misclassifications in 2025 were not randomly distributed but clustered in transitional neighbourhoods, of which many located along the western and south-central areas. The results shows that communities that are dynamic and marginal require additional model attention, potentially through higher temporal resolution data (monthly or weekly), integration of real-time social or mobility indicators, or spatially adaptive modelling techniques.

Overall, the results demonstrate that incorporating network-based livability, spatial dependency, and temporal trends produces a more spatially coherent and interpretable model of crime risk. At the same time, the limitations found point to key ways we can make GeoAI models more responsive to fast changes in cities.

Chapter 5: Discussion

Introduction

This chapter explains the findings from chapter 4 and situates them within the broader theoretical and methodological context of spatial criminology, GeoAI, and urban livability research. Whereas chapter 4 focused on reporting results, this chapter examines their meaning; how the spatial structure of crime in Chicago interacts with livability and accessibility; why the Enhanced Random Forest model worked better than others, and what these results mean for predictive modeling, city planning, and future research.

The discussion is not just a reiteration of the findings of Chapter 4 but instead it attempts to figure out what the findings entail, an alternative activity.

5.1 The Spatial Structure of Crime in Chicago

5.1.1 Persistent Spatial Dependence and Crime Clustering

The I results of the Moran created something that spatial criminologists would anticipate but one that would be worth investigating further; Chicago does not merely have a clustered crime pattern but a clustered pattern with an I ranging between 0.31 and 0.34 that is being held between six years. The fact that there is stability is more analytically relevant than the very clustering. It indicates that the geographic clustering of crime in Chicago is not merely a transient phenomenon, but is a feature of structural conditions of particular communities that are reproduced year after year.

An analogous argument is presented by the law of crime concentration by Weisburd (2015), which focuses on a smaller spatial level that the few micro-places represent a disproportionate number of criminal events. This community-level analysis is at a coarser scale, but the dynamic underlying it is the same crime is not evenly distributed, and the nonlinearity is not by chance. The west and the South side neighborhoods that seem to be continuous high-high in the local spatial analysis, Englewood, West Garfield Park, North Lawndale have featured in criminological works on Chicago decades ago. The social disorganisation thesis presented by Shaw and McKay (1942) pointed to structural disadvantage and undermined informal social control as the processes which maintained high levels of crime in exactly this type of community and later, Sampson (2012) research on neighbourhood effects demonstrated how

such circumstances perpetuate themselves across generations. The I result of the Moran are in a way a modern statistical replica of those results.

The spatial lag contribution to the model performance 6% of overall Gini importance that relates to the studies by Anselin (2003) on the spatial spillovers. The risk of crime in a particular community is partially dependent on what is occurring in neighboring communities, a fact that traditional independent-observations models do not take into account. The addition of the spatial lag is not a technical flourish: it is in reaction to an established regularity in the empirical literature that the diagnostics in this research validate.

5.1.2 Spatial Non-Stationarity and Local Variation

The GWR analysis introduced a complication that the global model performance metrics do not capture: the relationship between livability and crime is not uniform across the city. For example, improved accessibility to parks and healthcare had stronger crime-reducing associations in higher-income North Side communities, whereas similar amenities in areas of socio-economic distress showed weaker or more ambiguous effects.

This spatial variation is consistent with crime pattern theory, which emphasizes that opportunity structures emerge from the interaction between offenders' routine activities and the urban backcloth (Brantingham & Brantingham, 1993). It also reflects the idea that the same amenity environment can have different implications depending on local guardianship and place management conditions (Cohen & Felson, 1979; Cozens & Love, 2015). The observed spatial non-stationarity therefore supports the use of spatial diagnostics and locally sensitive analysis tools (e.g., GWR-style reasoning) to complement global models when interpreting livability–crime relationships (Fotheringham et al., 2002).

5.2 Livability as a Spatial Determinant of Crime

5.2.1 Livability Indices and Their Predictive Role

A central objective of this research was to evaluate the role of livability operationalized through the Unified Livability Index (ULI) and the network-based Distance-Based Livability Index (DBLI) as a predictive determinant of crime. Chapter 4 demonstrated that both indices displayed negative correlations with crime, with DBLI exhibiting a strong relationship ($r \approx -0.61$) and ULI a moderate one ($r \approx -0.48$).

These findings suggest that accessibility to essential amenities captured here through distance-based measures to services such as education, healthcare, commerce, and public transport provides a meaningful contextual signal when analysing community-level crime patterns. Accessibility is a core dimension of livability and well-being frameworks and is commonly operationalised through access-to-services and walkability indicators (Pacione, 2003; OECD, 2020; U.S. Environmental Protection Agency, 2021).

Rossmo's (2000) work on geographic profiling provides one theoretical anchor for DBLI specifically, travel costs and proximity shape where offenders operate and where targets are vulnerable, even when individual journeys are not directly observed. A measure of how easily residents can move through their neighbourhood via the pedestrian and transit network speaks to the same structural conditions from the other side which is the guardianship and activity patterns of non-offenders.

5.2.2 Synergy between Amenity Density and Accessibility

The ablation study's finding that neither ULI nor DBLI improves the model independently, but their combination does, is the result of this section that deserves the most attention. It would have been convenient if one index had proven sufficient, that is, computationally simpler and easier to communicate. The fact that both are needed tells us something about what livability actually is.

A neighbourhood where services are densely mapped but difficult to reach via the real network, perhaps because of physical barriers, limited transit, or discontinuous street layouts, scores well on ULI and poorly on DBLI. The reverse configuration is also possible, with reasonable network accessibility to a sparse distribution of services. Both cases are not synonymous with true livability, and the model seems to be sensitive to the difference in a variety of ways that cannot be reflected solely by index. This aligns with urban analytics models that consider livability as irreducibly multidimensional, a composite of environmental quality, accessibility, and connectivity and not as any one dimension (Pacione, 2003; OECD, 2020).

5.3 Interpretation of Machine Learning Performance

5.3.1 Why the Enhanced Random Forest Model Performed Best

The performance boost of the Enhanced Random Forest as compared to the Logistic Regression is easy to understand, the interrelation between livability and spatial context and the risk of

crime are not linear and a regression that is linear will fail to capture them. The fixed coefficients of logistic regression cannot capture crime thresholds, interaction between amenity accessibility and historical crime concentration, or the spatially varying relevance of various features. These complexities are accommodated by random Forest which is a collection of decision trees, each constructed using a bootstrapped sample of the data and random feature selection, which the analyst does not need to pre-specify (Breiman, 2001).

The question that is more interesting to consider is why XGBoost that also addresses non-linearity did not improve even with the improved feature set. A sensible understanding is that the regularization structure of XGBoost had already acquired a notion of approximating some of the interactions that ULI, DBLI, and the spatial lag capture. When the model has successfully learnt the patterns of the neighbourhoods with the help of the gradient boosting steps that it takes, introducing features that capture a similar underlying structure to some degree might not alter the outputs of the model. This is purely speculative since no additional ablation work has been done, but implies that the role of livability features could be contingent on whether the base model is already reproducing the corresponding spatial structure in other ways.

5.3.2 Temporal Stability and the Challenge of Predicting Emerging Hotspots

The drop in performance in 2025 is not merely a failure but a diagnosis. The 20 false negatives of that year did not evenly spread throughout Chicago; they were clumped together in communities with long-term crime patterns that disproportionately rose above the hotspot threshold at a rate exceeding what the temporal lag characteristics could predict. It is more helpful to know why the model did not work in those particular communities than the overall F1-score, since it shows in which situations the assumptions of the framework may not hold.

The assumptions should be explicitly named. The Enhanced Random Forest is based on the following assumptions: recent crime history is the key predictor of near-term risk, neighbouring community conditions offer consistent contextual data, and livability, an annual measure based on relatively stable data sources, is the environmental conditions of interest in crime. The three assumptions are fairly well-constructed when it comes to communities in stable trajectories. All of them augur poorly in terms of communities that have undergone rapid socioeconomic transformation, have suddenly changed policies or concentrated transient events that are smoothed out by annual aggregation.

This is an established issue in the predictive policing literature (Kounadi et al., 2020; Du and Ding, 2023), yet the 2025 outcomes provide it with a certain geographic nature. The

communities which the model overshot are not chosen randomly in Chicago; they are the transitional communities, the ones between the persistent hotspot and persistent low-crime areas, and they are where structural characteristics such as livability and spatial lag have the least useful information about where the travel is headed currently. That trend suggests what would most contribute to such a framework would be complementary data, monthly crime rates, mobility measures, and economic stress indicators.

5.4 Implications for Urban Planning and Policy

5.4.1 Preventive Planning Through Livability Interventions

This combination of the DBLI-crime relationship and the contribution of the index to the model performance indicates something that has long been argued by urban planners and researchers in the field of public health: structural conditions, the reachability and accessibility of certain services, and their ease of access are not just background conditions to crime, but determinants of crime. This is not directly translated into a policy prescription as correlation is not causation and GWR results demonstrate that the same amenity conditions yield different results in different neighbourhoods. Nevertheless, the argument suggests interventions like:

- Improving pedestrian and transit connectivity
- Addressing “amenity deserts” in disinvested communities
- Expanding healthcare, education, and recreation infrastructure
- Enhancing mobility networks that reduce isolation

Such interventions align with public health and urban equity frameworks, offering an alternative to enforcement-based approaches.

5.4.2 Open Data and the Scalability of GeoAI

A feasible aspect of this research that can be mentioned is that the whole pipeline of analysis is performed with the help of publicly available data and free-of-charge tools. Open crime records, OpenStreetMap amenity data retrieved using the OSMnx, spatial weights created using PySAL and machine learning using Scikit-Learn, all of which are free, and none of them need institutional access or proprietary systems. The city that is similar in terms of open data infrastructure could replicate or modify this framework without expert resources using a municipal analyst.

A common criticism of GeoAI methodologies of crime prediction is that they are designed in research environments and cannot be applied to operational environments without major technical investments. The workflow in this case is created in opposition to that tendency but to create a purposefully simple, clear workflow and based on the tools that are already widely adopted in the field.

5.5 Methodological Reflections and Limitations

5.5.1 Strengths of the Approach

The decision to use network distances as opposed to Euclidean proximity in DBLI was a substantive decision, not a technical one. The straight-line distance between a community centroid and the closest hospital does not inform us much about the ability of the residents to reach that hospital. Even at the community level, road and pedestrian network distances give a more true-to-life view of accessibility, and the results of the model using the augmented feature set indicate that this can be important in prediction.

The choice to use spatial diagnostics at each stage such as autocorrelation like testing before modelling, spatial lag effects and analysis of residuals to show any remaining spatial structure ensured that the modelling options were not forced to the data but were based on its actual form. The rolling backtest design also indicated a promise to test the model as it would be used in practice, trained on the past and tested on the future, as opposed to taking advantage of random cross-validation splits that would boost the apparent performance.

5.5.2 Limitations and Areas for Improvement

The most important constraint is annual aggregation which is not abstract but is concrete with the 2025 results. A framework which is unable to identify the emergence of hotspots within the year of their emergence, is limited in its operational utility, however well it works within well-known conditions. This would not completely address the issue, since monthly or quarterly data would reduce the lag between a change in the condition of a community and the model being able to notice it.

A similar issue is with the spatial scale. The community areas consist of 77 good units to work with in an analysis of the city, although it covers up variation that is important to comprehend crime. Two communities with similar annual crime counts might have very different internal geographies: one concentrated around a single corridor, the other distributed more evenly and

the community-level model treats them identically. Street-segment level analysis, as Weisburd et al. (2016) demonstrate, reveals patterns that community-area analysis systematically obscures.

The livability indices measure presence and network accessibility but say nothing about quality.

Finally, the model has no mechanism for capturing what might be called socio-political shocks, that is, changes in policing strategy, economic disruptions, housing market shifts, or community-level events that alter crime risk without leaving a prior trace in the annual data. The 2025 false negatives almost certainly reflect some combination of these factors. Until the model can incorporate real-time or near-real-time signals of this kind, its sensitivity to rapid change will remain limited.

Chapter 6: Conclusion and Recommendations

Introduction

Research projects of this kind tend to produce two kinds of conclusions: the ones is set out to find, and the ones that is actually found. This study was designed to test whether spatially derived livability measures operationalized through the Unified Livability Index and the Distance-Based Livability Index could improve crime hotspot prediction beyond what historical incident data alone provides. That question was answered, reasonably clearly, in the affirmative. But the more instructive result may be the 2025 performance drop, which the study was not designed to produce and which turns out to be analytically richer than the F1-scores that preceded it.

This chapter draws together what the research established, what it contributed, where it fell short, and what the next step would look like. It does not attempt to overstate what a community-level, annually aggregated, open-data framework can achieve, that would misrepresent both the findings and their limits.

6.1 Summary of Key Findings

6.1.1 Spatial Dependence and Persistent Hotspots

The Moran's I results across 2020 - 2025 confirmed what criminological research on Chicago has documented for decades, that crime does not distribute itself randomly across the city's 77 community areas. High-crime communities cluster together, and they do so consistently, year after year, with values between 0.31 and 0.34 that show no sign of dissipating. West Garfield Park, Englewood, and North Lawndale appear as persistent high-high clusters not because of anything the model is doing, but because the structural conditions that concentrate crime there, documented through social disorganisation theory (Shaw & McKay, 1942) and Sampson's (2012) neighbourhood effects research, have not materially changed.

What this means for the modelling is straightforward: treating Chicago's 77 community areas as independent observations is not defensible, and the spatial lag feature is not an optional enhancement. It is a methodological response to a structural property of the data. That the spatial lag contributed 6% of the Random Forest's Gini importance is consistent with what the spatial diagnostics predicted it would do.

6.1.2 Livability and Accessibility as Predictive Determinants

The negative correlation of the DBLI with crime ($r \approx -0.61$) is the negative news, but the more valuable result is the ablation study result that neither ULI nor DBLI was better than the model alone, but their combination was. That difference took a while to value in the analysis, and has a practical implication which a single-index method would have overlooked.

The combination of contribution discloses that density and accessibility are assessing different aspects regarding the same neighbourhoods. A community with its services in place, but relatively inaccessible through the pedestrian and transit network is not synonymous with a community where the services are actually accessible. The difference is sensitive to the model in such a way that a single composite livability score would not be. It is a minor result in the terms of F1-score change, but the real implication of it in the context of the livability operationalisation in urban planning practice is that collapsing density and accessibility into one index is a common practice.

6.1.3 Predictive Performance of Machine Learning Models

The Enhanced Random Forest was tested on the 2023 - 2024 test data with $F1 = 0.967$ and $ROC-AUC = 0.997$, and the rolling backtest showed that the performance of the model was consistent across training windows. These are good results and they were repeated across all three classifiers when livability and spatial features were added which indicates that the improvement is an attribute of the features as opposed to a specific algorithm.

It is at the 2025 outcome where the analysis ends up being more intriguing than the headline measures would indicate. There were 20 false negatives communities that turned into hotspots but were not forecasted to be so were located in the transitional belt between the South and the West Sides of Chicago, which persistently had high crime levels, and the more steady communities. The model had not learned to anticipate rapid change in those communities because their recent history gave it no reason to. Annual temporal lag features project persistence forward; they have no mechanism for detecting acceleration.

6.1.4 Spatial Realism and Modelling Limitations

It would be easy to treat the 2025 performance drop as a limitation and move on. The more productive framing is to treat it as a finding that defines the boundary of what this type of model

can and cannot do. Annual aggregate frameworks built on structural features crime history, livability indices, spatial lag are good at capturing persistent, slowly-evolving risk. They are not well-suited to detecting communities in transition, where conditions shift faster than the data can track.

That boundary is not unique to this study. It is an attribute of the modelling approach, it is easier to call it that than to ignore it or consider it a disqualifying flaw.

6.2 Contributions to Knowledge

6.2.1 Theoretical Contributions

The work is not a new criminological theory, but it is an empirical finding that there is a relationship that the literature has hypothesized more than it has been tested: that urban livability, specifically the network-based accessibility of essential services is a spatially measurable correlate of crime risk that can add predictive information to that which is already available in terms of historical incidents. It is easy to retrospectively ground this in the Routine Activity Theory (Cohen and Felson, 1979) and social disorganisation frameworks (Shaw and McKay, 1942) but the empirical evidence on the community level using open data and reproducible methods adds something to theoretical connections that cannot be demonstrated theoretical argument alone.

The GWR findings also introduce a certain level of complexity: the correlation between livability and crime does not exist throughout the city, so hypothetical arguments about amenity accessibility and guardianship should not be made at the global scale. Methods of place-based and routine activity paradigms have been found to be more spatially sensitive, and place-based analyses have shown that the same park or transit stop has various implications in different neighbourhood contexts

6.2.2 Methodological Contributions

The DBLI is the study's most concrete methodological contribution. Using OSMnx-derived pedestrian and road networks to calculate actual travel distances between community representative points and essential amenity categories produces a fundamentally different measure from Euclidean proximity one that reflects how people actually move through cities rather than how a crow would fly. The fact that DBLI outperformed ULI in its correlation with

crime, and that the two together produced model improvements that neither produced alone, provides post-hoc validation for the construction choice.

The broader workflow spatial diagnostics informing feature engineering, temporal splitting enforcing realistic evaluation, ablation testing isolating feature contributions is designed to be reproducible using publicly available tools and data. Whether the specific Chicago findings generalise to other cities is an open question, but the pipeline itself is transferable.

6.2.3 Empirical and Policy Contributions

The most practical input is the most direct, a framework which municipal analysts can tailor without proprietary data or specialist infrastructure. OpenStreetMap, OSMnx, PySAL and Scikit-Learn are all free and open crime records of Chicago. A similar city with open data might use a variation of this pipeline and employ it to locate communities where a lack of access to amenities and a high rate of crime persistence is indicative of a need to invest in infrastructure not to replace community knowledge, but to complement it.

This is better accomplished using the risk probability surface as opposed to the binary hotspot classification. It conveys the sense of uncertainty and gradation in a manner that a binary label does not, important when outputs are being utilized to make resource allocation decisions with actual implications to actual communities.

6.3 Policy and Planning Implications

6.3.1 Data-Informed Crime Prevention

The DBLI results affirm a planning argument that public health and urban equity researchers have long been making, but that crime prediction research has been slower to consider structural circumstances that not only define which services are accessible, through what pathways and in what timeframes, are not just contextual backdrop to crime but are active determinants of it. Increasing transit connectivity, sealing gaps in provision of services in the disinvested communities and making investments in pedestrian infrastructure in the regions where DBLI found accessibility-poor are not merely quality-of-life strategies. The correlation pattern of this data indicates that they are crime prevention strategies as well over time.

Care should be taken in making this argument. The research creates correlation, but not causation and the GWR findings indicate that the livability-crime relationship is not the same

throughout the city and cannot be easily generalised. However, the communities with lowest DBLI and with the greatest risk of crime are quite similar communities that have had the longest history of disinvestment. It may help to frame sustained investment in infrastructure in those places as crime prevention (not charity or social policy) so that it can be more competitive in competing against resources in planning environments where crime prevention is a more important political priority.

6.3.2 Limits of Algorithm Prediction

The 2025 false negatives is a reminder that predictive models are not an oracle, but a tool. The communities the model overlooked in 2025 did not become hotspots due to the failure of the algorithm to learn on the data; they became hotspots due to the conditions changing in such a way that the annual data would not be able to capture them in time. The local conditions, real-time observation and relationship between human knowledge of the local conditions, and the relationship between the community are also an important complement to any risk assessment based on the models.

The implementation of such a framework has to be ethical, which means it has to be clear what the model is able to observe or not. Uncertainty and the anticipated risk should be communicated using risk maps, and communities that are identified by human analysts as areas of concern, although the model has not yet caught up, ought to count in resource allocation decisions. When the results of the algorithms are perceived as authoritative and not advisor-y, there is a danger of focusing punitive efforts on already disadvantaged communities without providing the community investment that the livability findings would lead to improving the structural conditions.

6.4 Limitations of the Study

The community area scale was a practical one; it is the scale at which both crime and demographic data are available in a consistent manner across Chicago, yet it is also a practical limitation of the analysis. The processes of crime exist on the street segment level; the concentrations that Weisburd et al. (2016) define as the most important to prevent crime often are limited to several blocks. Moving towards the community boundaries removes this variation and renders the model insensitive to the micro-geographic dynamics that are most practiceable in policing and planning.

This is compounded by yearly aggregation. A community, in which crime increases sharply during the second half of a year, appears in the annual record as slightly different in appearance than last year. The sources of the 2025 errors are directly related to this: new hotspots that were developed during the year were not visible to the model until they were training data in the next cycle. This is not a defect with the existing framework, but a characteristic of the data resolution, but it specifies what the next version would have to do.

The livability indices are based on its existence and access, rather than its capacity or its real utilization. An open street map transit stop that is infrequently run or a park that is technically accessible but in poor condition does not add to the safety as the index suggests. The qualitative aspects of urban infrastructure, the aspects that distinguish between the services being actually available and merely being available, are more difficult to operationalise using open data, but their lack in the framework is a substantive constraint and not a minor caveat.

Last, but not least, the research is predictive and not causal. The associations between DBLI and crime, and the scores of feature importance of the Random Forest do not prove that a better availability of amenities would decrease crime. It would take designs that explicitly tackle confounding concerns, e.g., quasi-experimental designs, longitudinal causal inference designs, or natural experiments, that the current thesis cannot cover.

6.5 Recommendations for Future Research

The loss of performance in 2025 is most oriented towards the value of the higher-frequency data. Switching the yearly to monthly or quarterly counts of crime would reduce the delay between the conditions varying and the model being capable of noticing it. Mobility data, dynamic inputs and emergency service calls as well as economic stress indicators would also give further early warning to communities in transition. These are the communities that the model overlooked in 2025, where the signals that it would have most informed would have been, making them the natural point of departure in a more finely-resolution follow-up study.

The second obvious extension is a much finer spatial scale. The level of analysis (census tract or street-segment) would bring the framework into the same spatial resolution as crime concentration really functions, and would enable the livability indices to be built on a level of granularity at which the difference between accessible and inaccessible services might be analytically meaningful. The computational requirements at that scale are also very high, but the computational tools adopted here, the OSMnx and the PySAL, are well-adapted to it.

Replication across cities would answer this question that this one city study alone cannot: is the DBLI-crime relationship, and the interaction between density and accessibility indices, a Chicago specific phenomenon or is it indicative of more general information about how urban structure influences crime risk. Toronto, Philadelphia and New York are similar in the open data infrastructure and would be good places to start. Reproduction in cities of other urban forms, lower density, polycentric or fast developing cities would be more difficult but would be more theoretically enlightening.

On the model side, explicitly spatio-temporal models, like LSTMs, Temporal Convolutional Networks, and Spatio-Temporal Graph Neural Networks, might be compared to the Random Forest baseline to determine whether they can learn sequence-based temporal dependence and spatial propagation using networks provides an improvement over the current lag-feature model. The models would have to be very carefully designed to prevent exaggerated performance estimates and their interpretability would have to be handled using SHAP or post-hoc explanation tools based on LIME. However, when the aim is to identify new hotspots not just to identify persistent ones, architectures that can model communities as a dynamic network nodes would be a more natural fit as opposed to annually lagged tabular features.

6.6 Overall Conclusion

What this study aimed to demonstrate was that space context operationalized using livability indices constructed using open data contributes something to crime prediction not offered by historical incident data alone. The findings can substantiate that assertion, with a small but significant difference in the unfolding over the classifiers and across the years of the test. The Enhanced Random Forest is superior to the baseline and superior in aspects that can be traced back to particular features that are theoretically supported by the study of spatial criminology and urban accessibility.

The thing that the study did not attempt to demonstrate, but instead demonstrated regardless, is where such model reaches the end of the road. Persistence is good at being projected by annual structural features. They do not foresee change. The communities that became hotspots in 2025 without the model predicting them are not a failure of the framework they are a signal about what the framework is and what it would need to become to be genuinely useful in the fast-moving conditions that characterise urban crime in transitional neighbourhoods

Crime is a spatially embedded social phenomenon. Predicting it well requires not just better algorithms but data that is temporally finer, spatially more granular, and sensitive to the

structural conditions access, infrastructure, opportunity that shape whether a neighbourhood is moving toward stability or away from it. This study is a step in that direction, not a destination.

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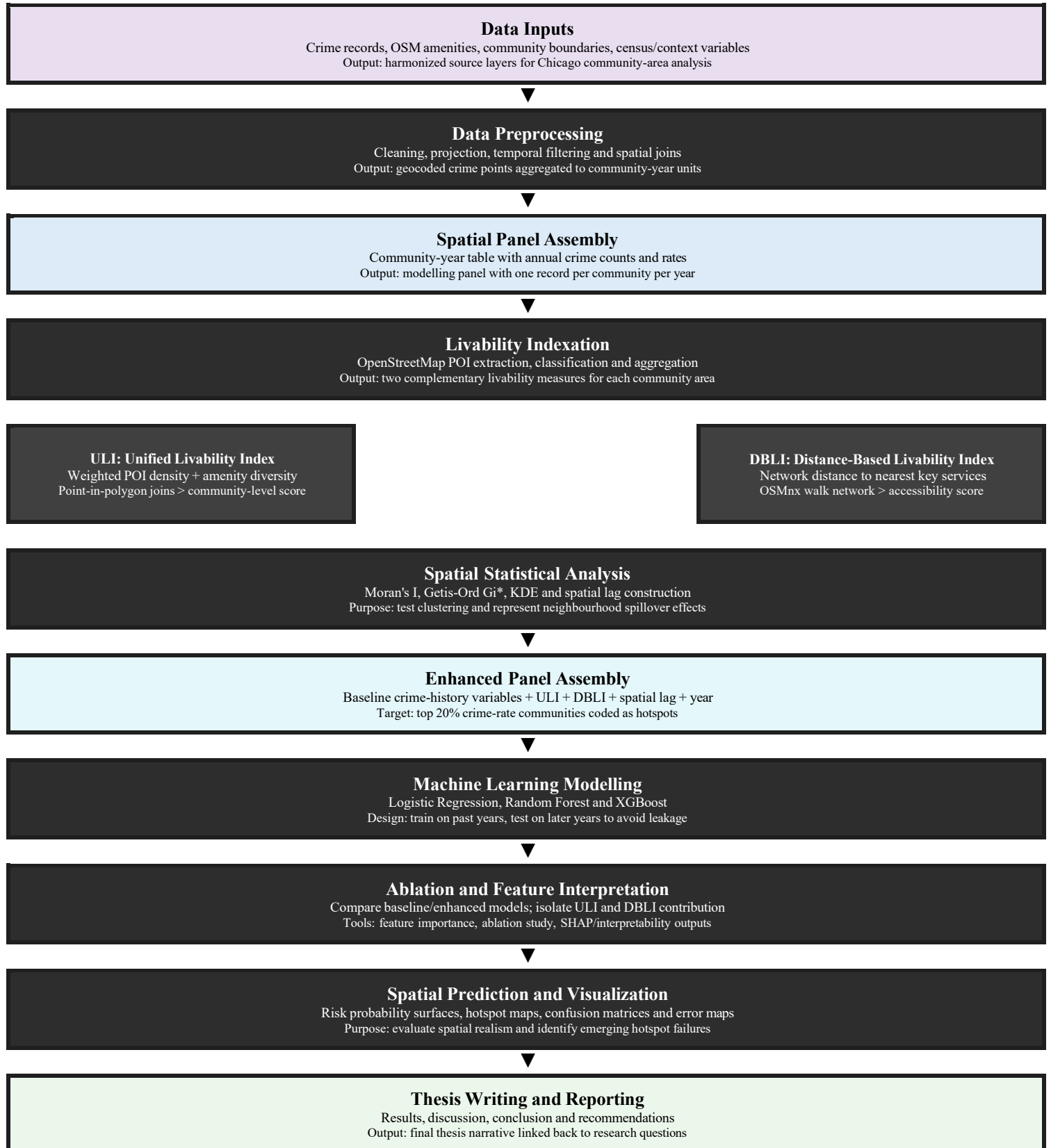
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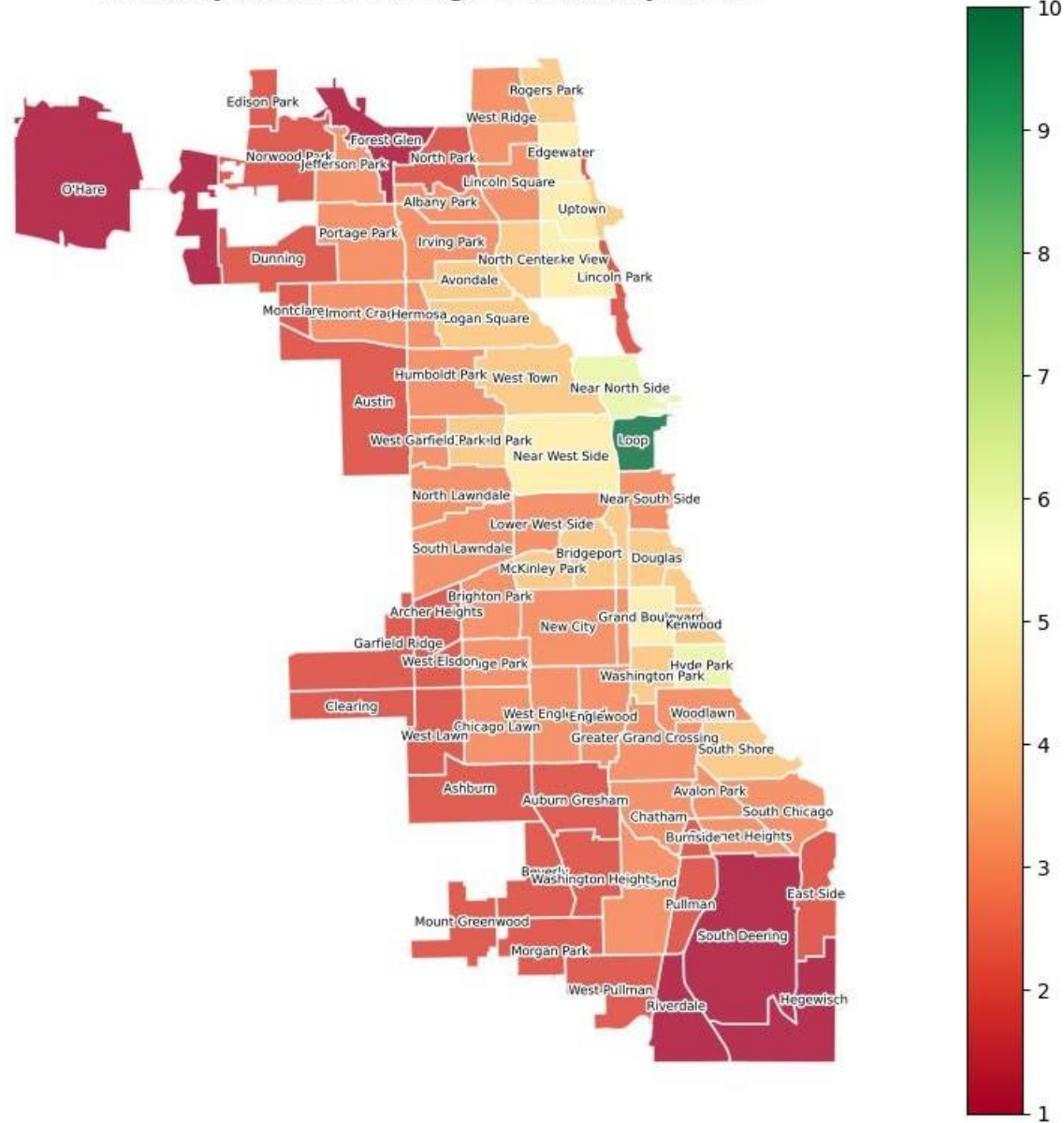
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Appendix

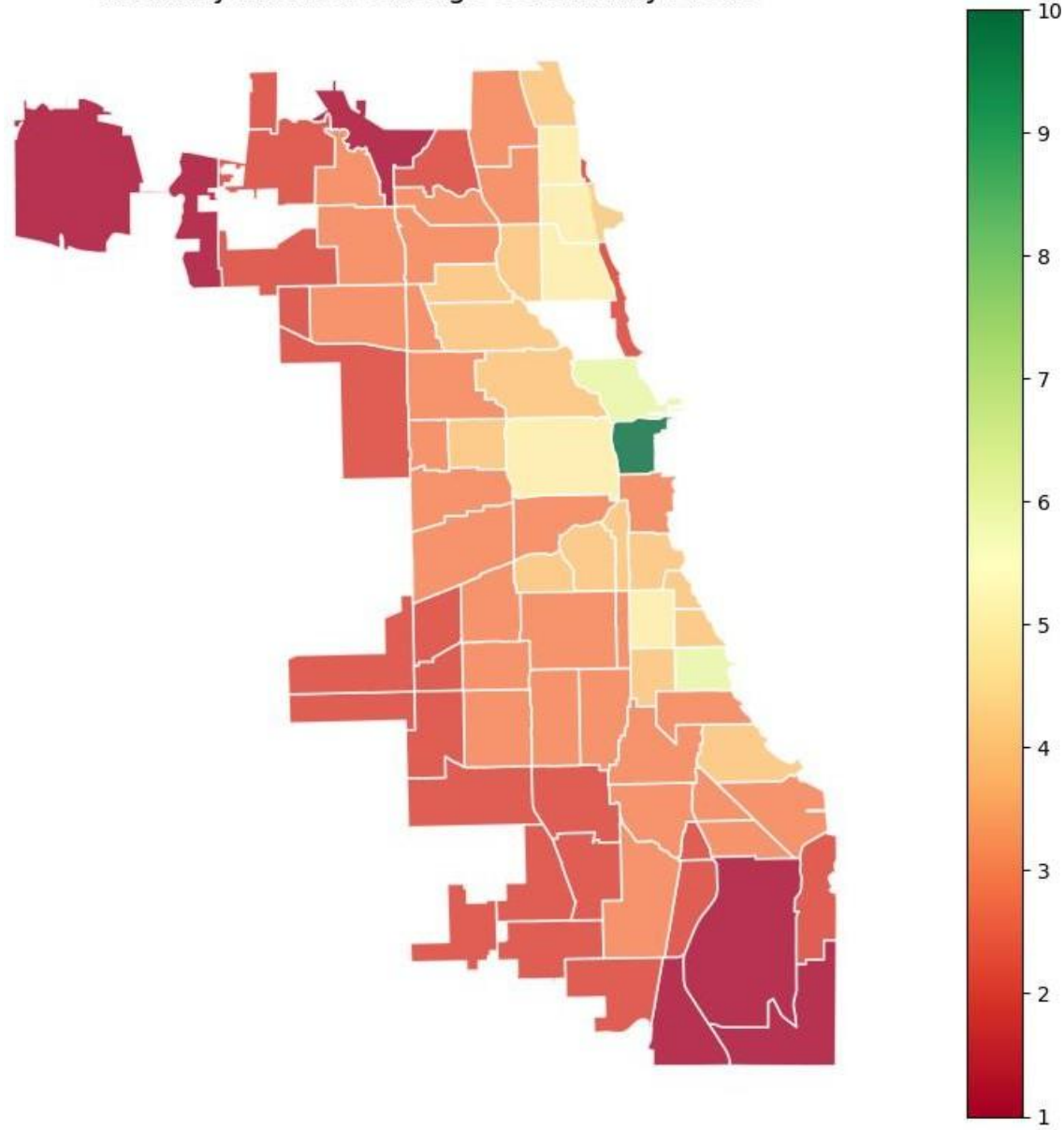
Detailed Methodological Workflow for Crime Hotspot Prediction

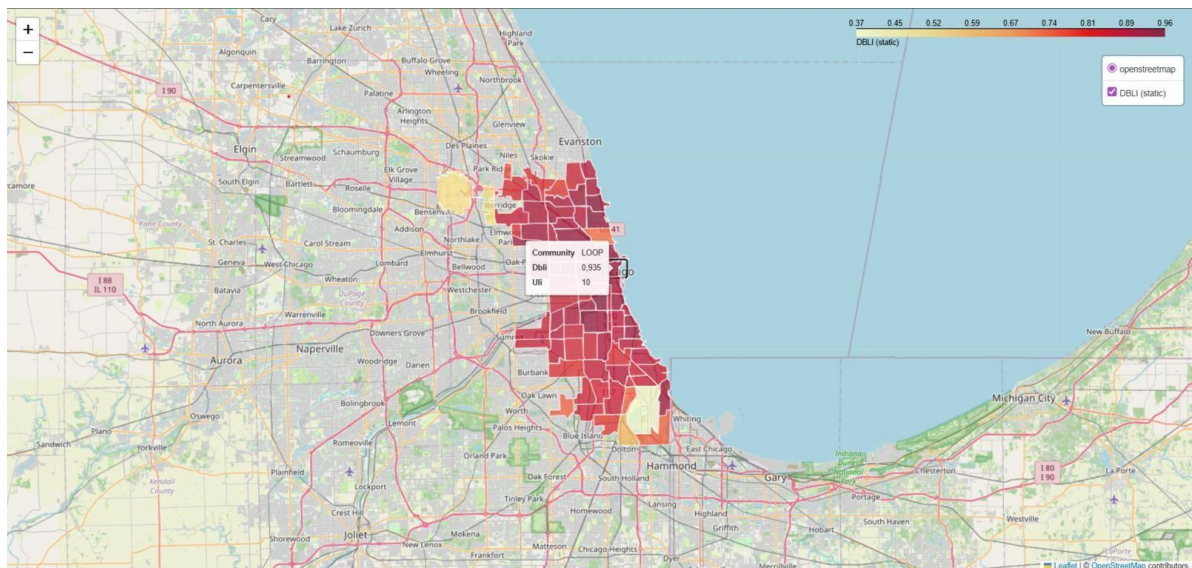
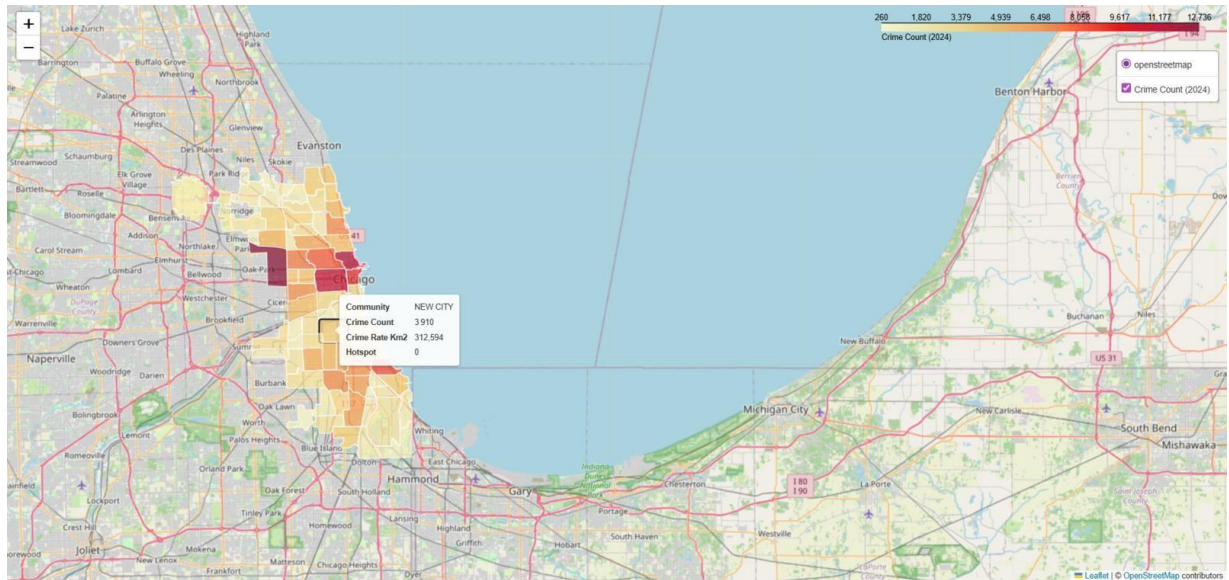


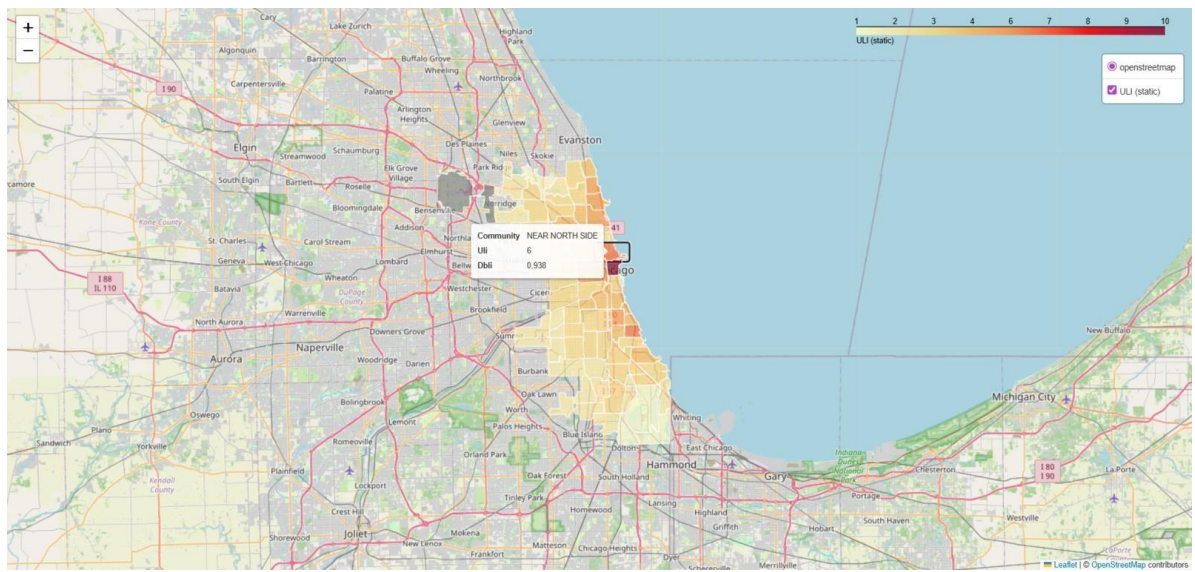
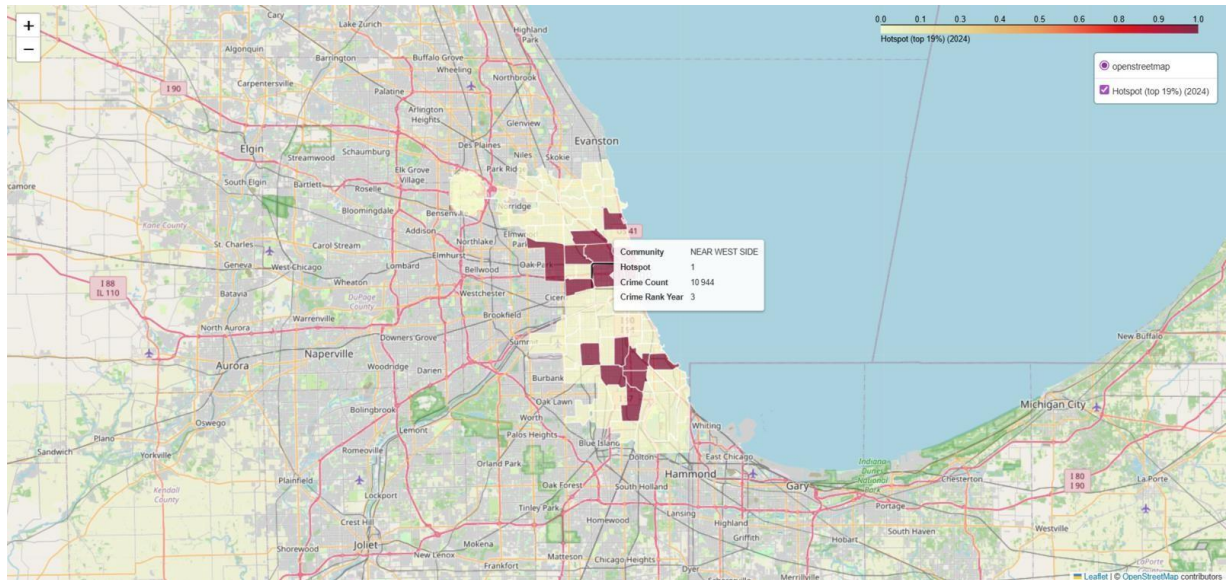
Livability Index of Chicago Community Areas

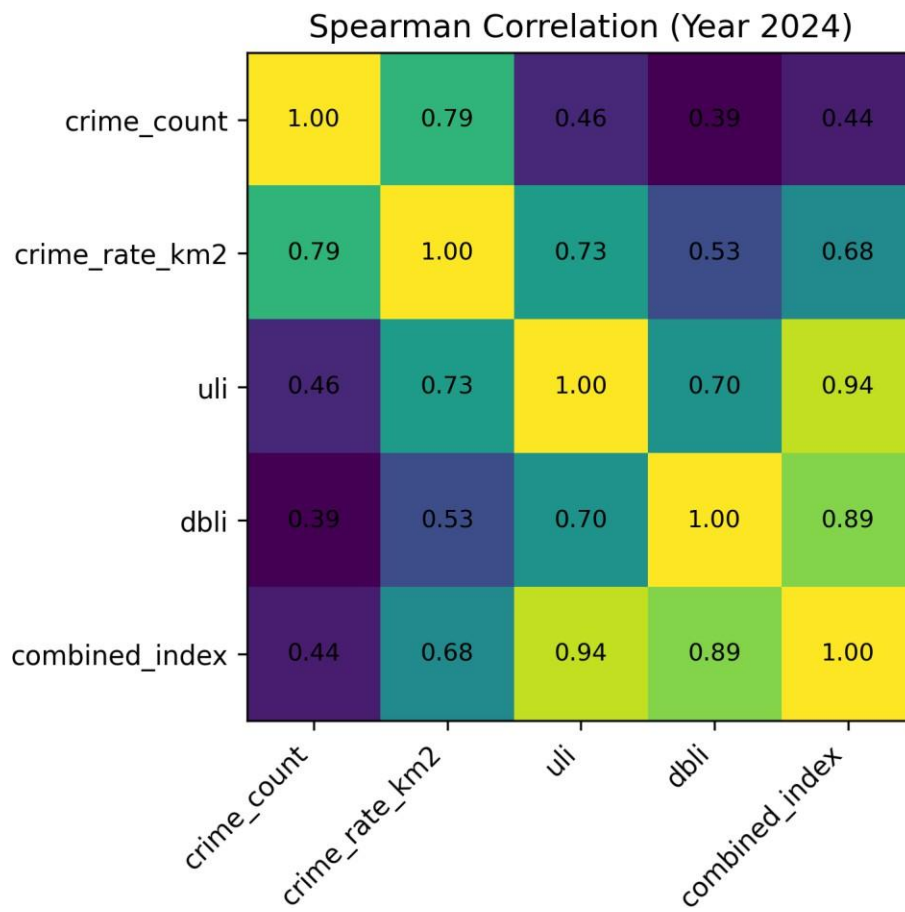


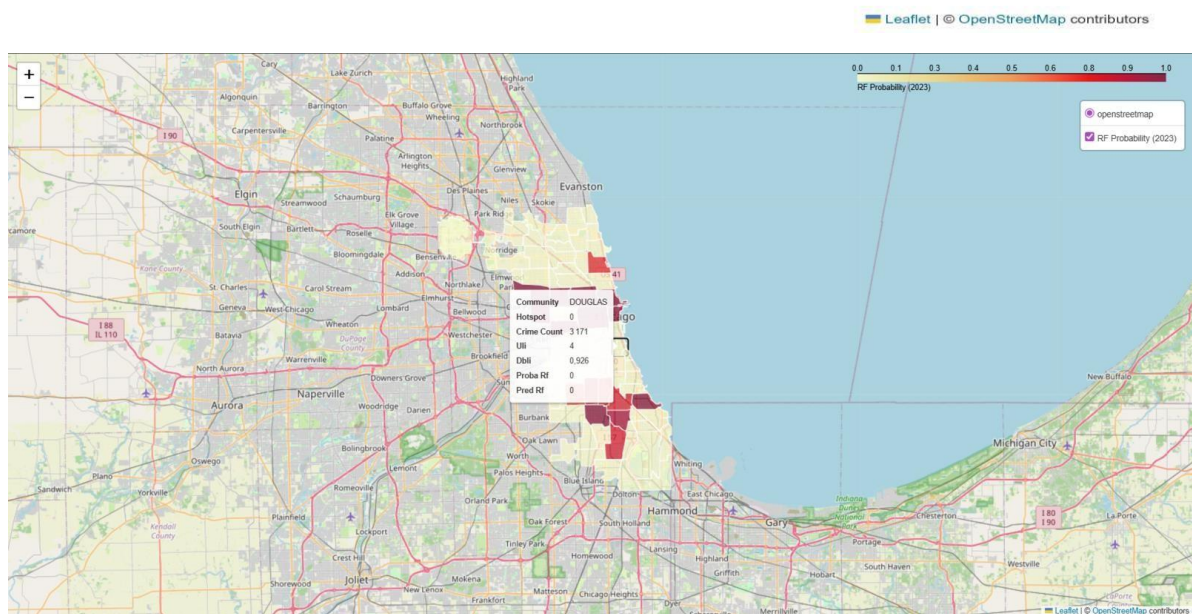
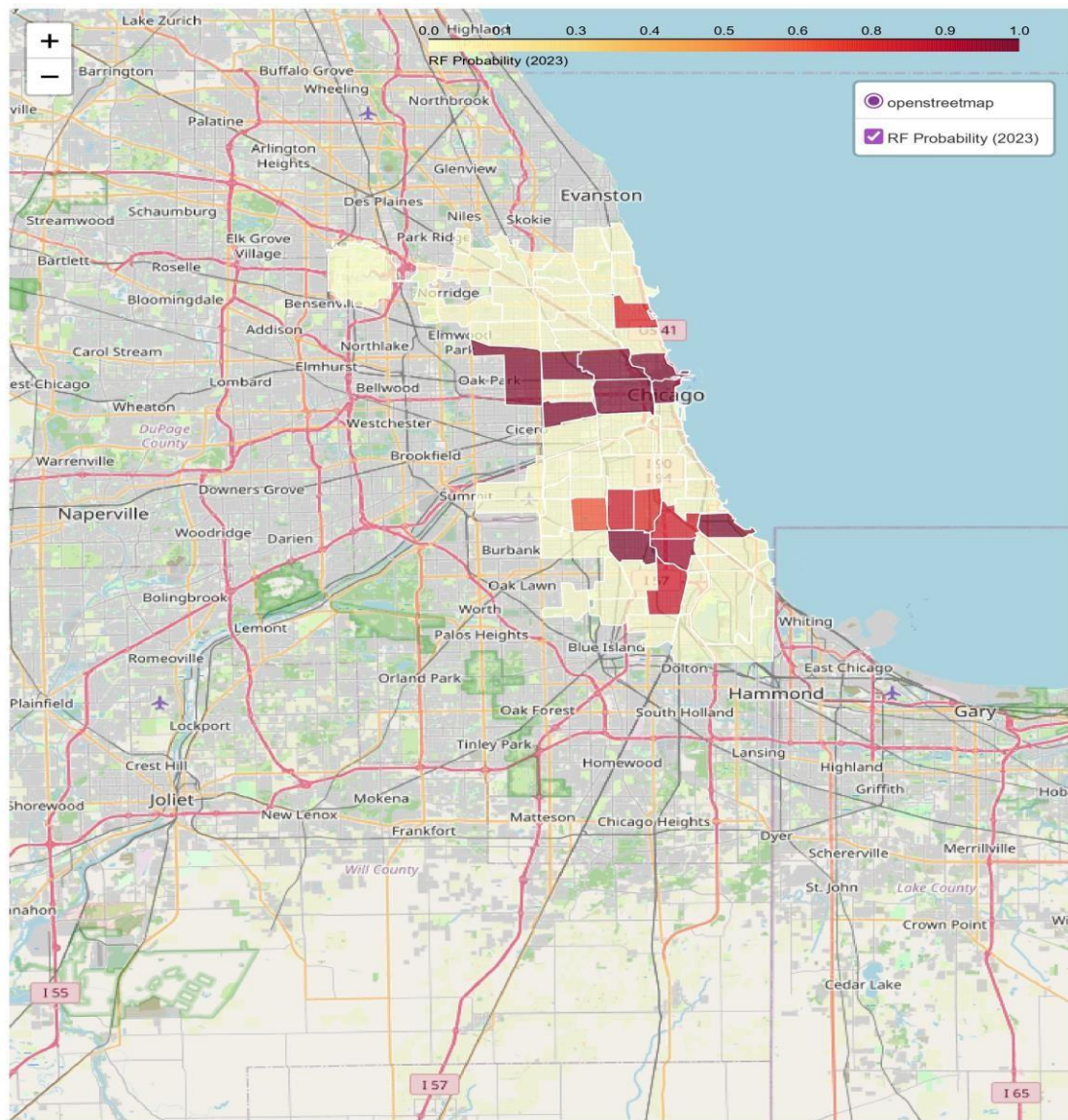
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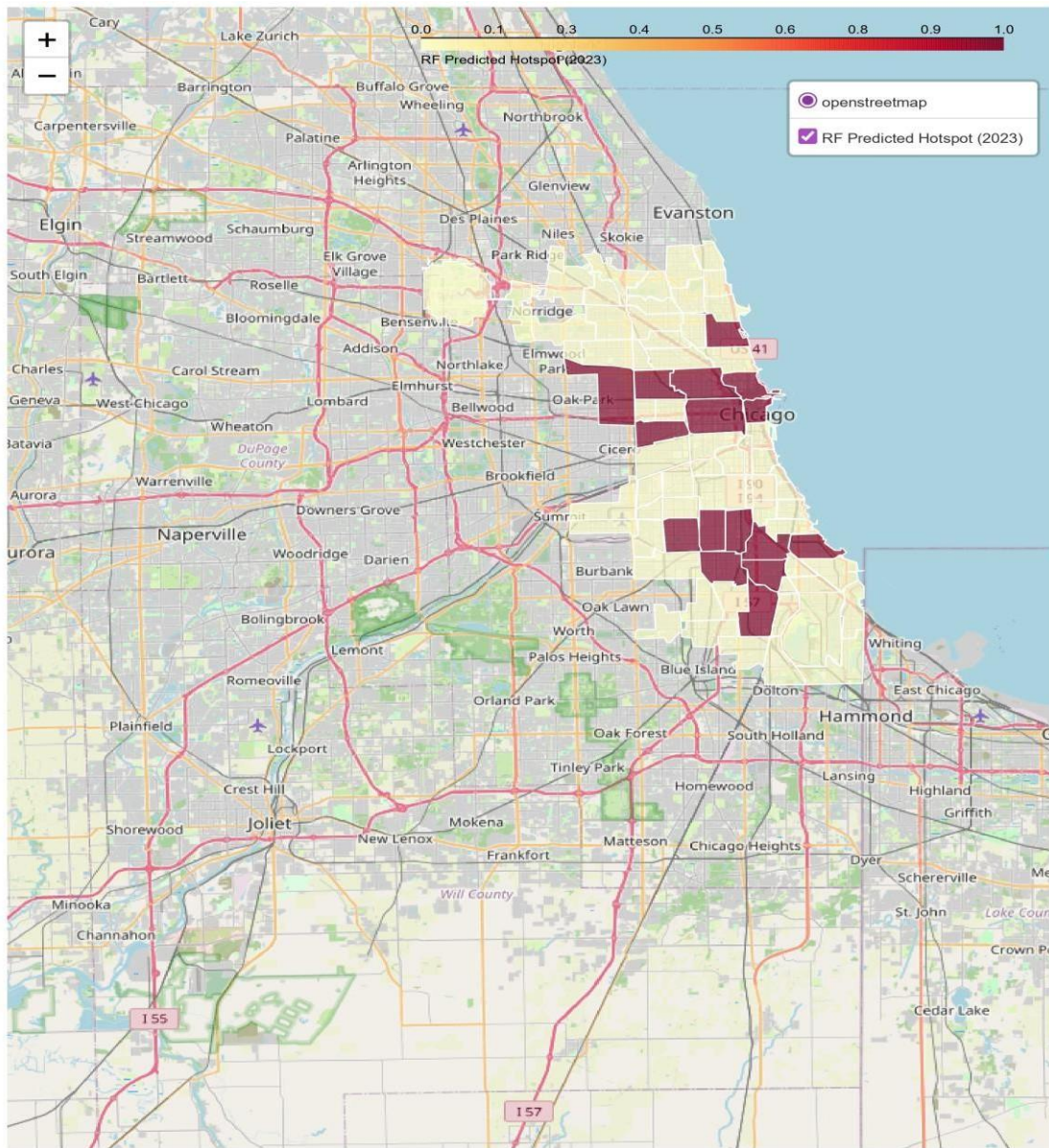




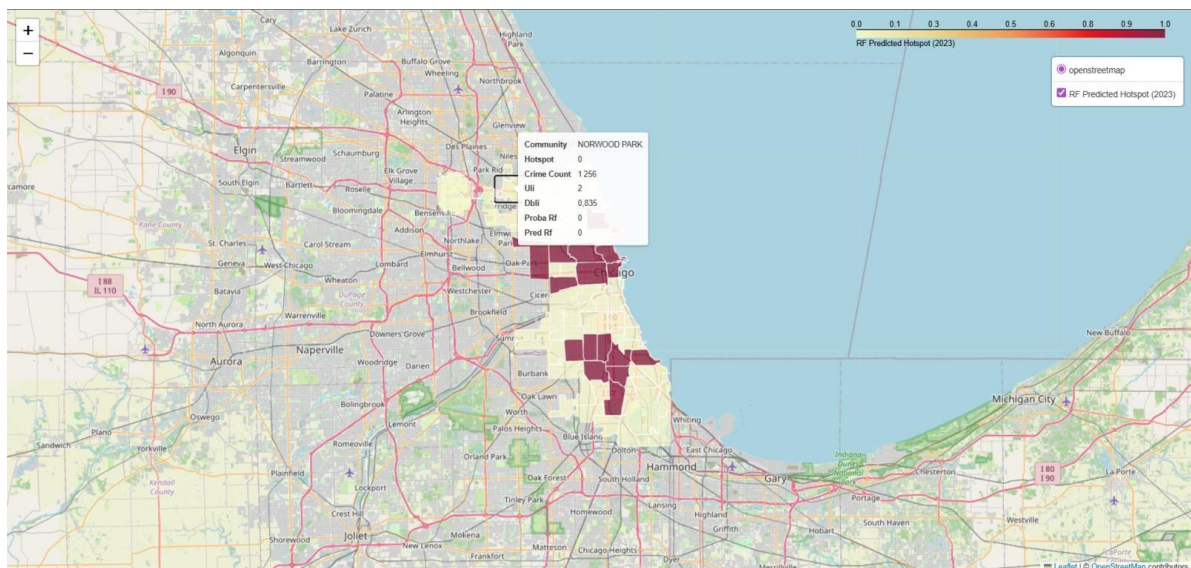


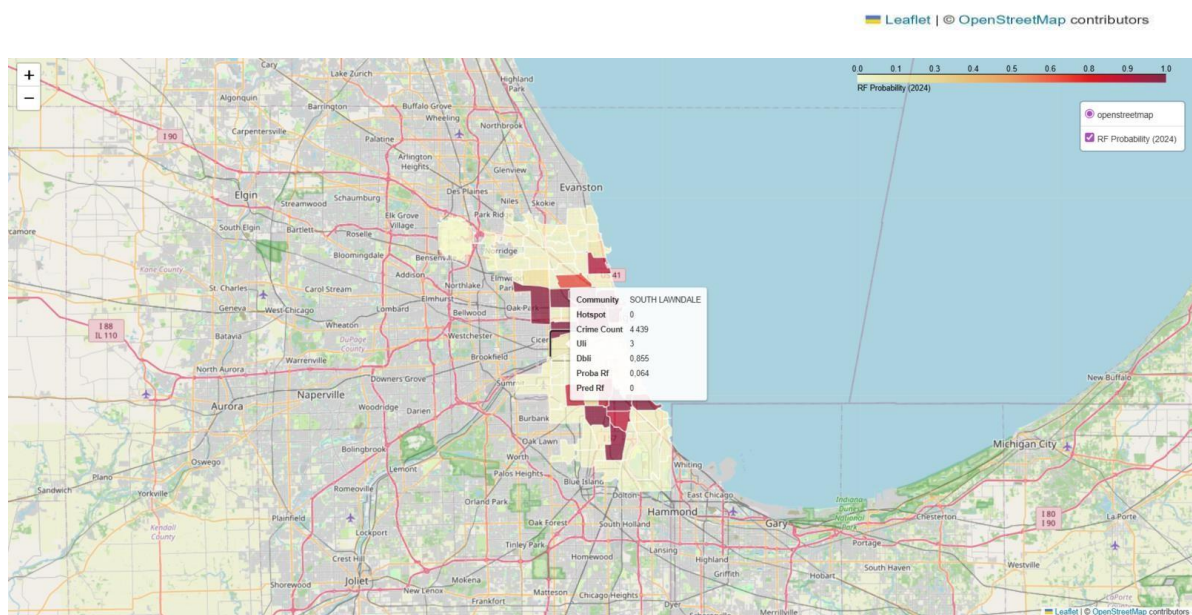
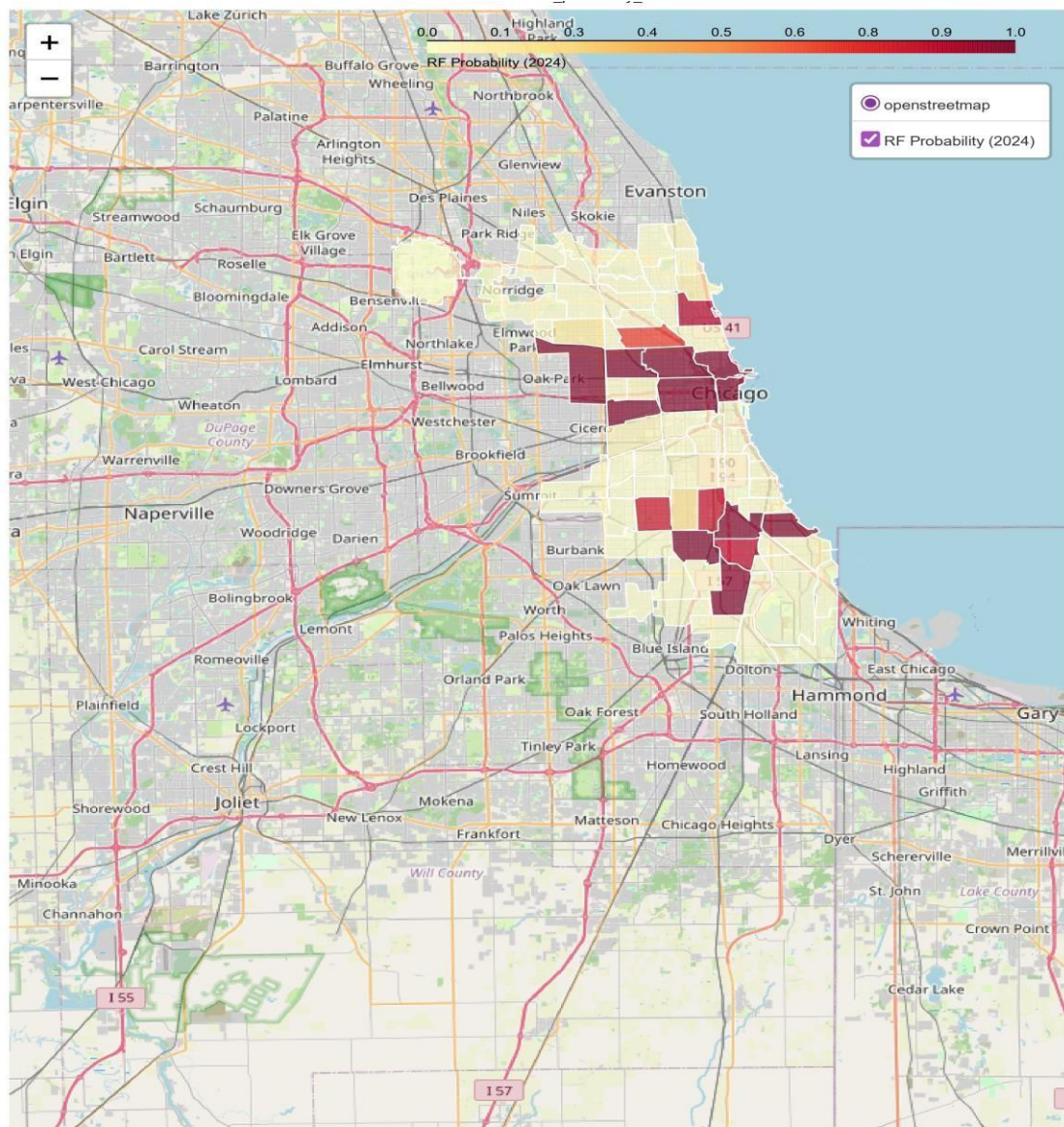


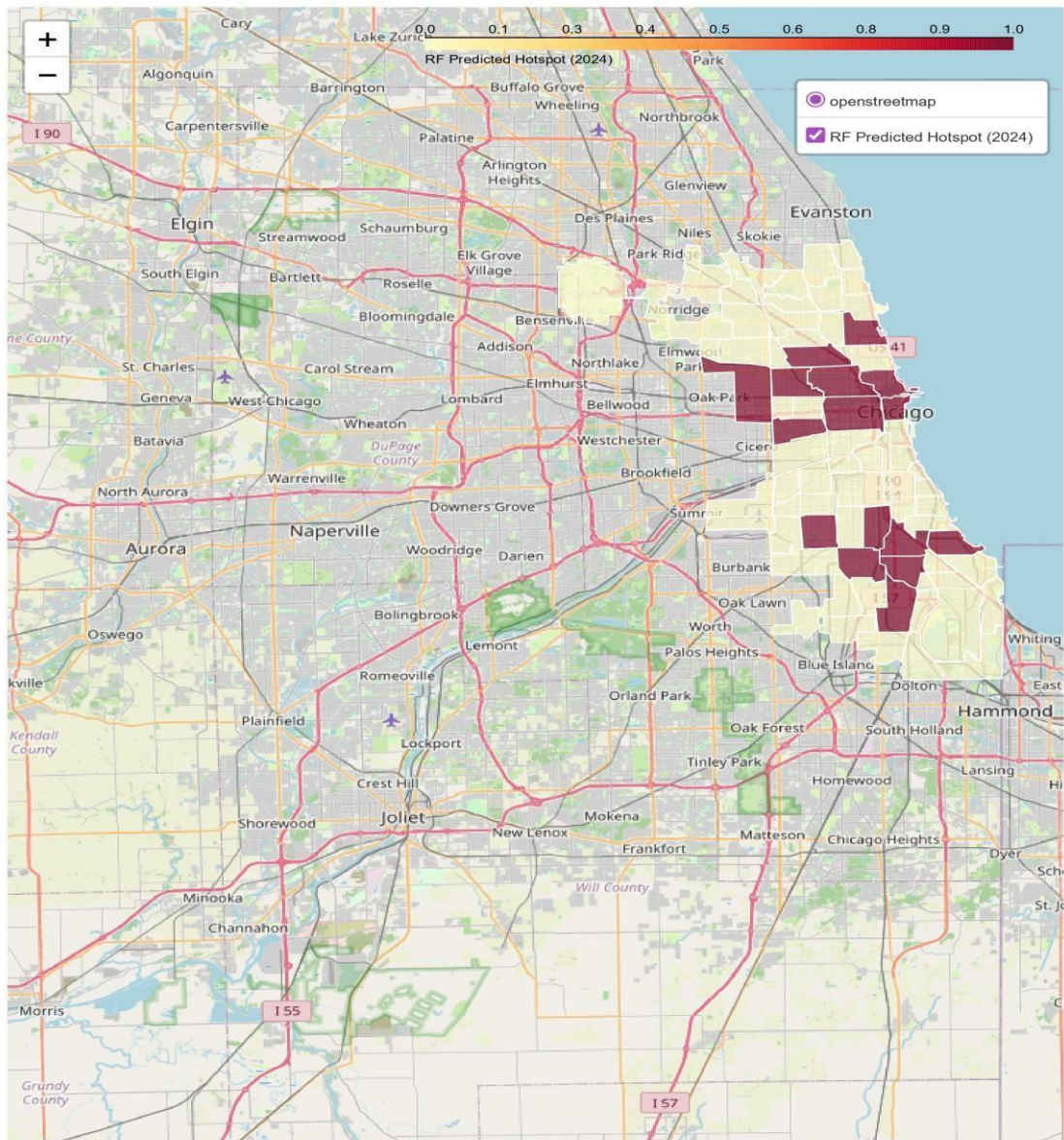




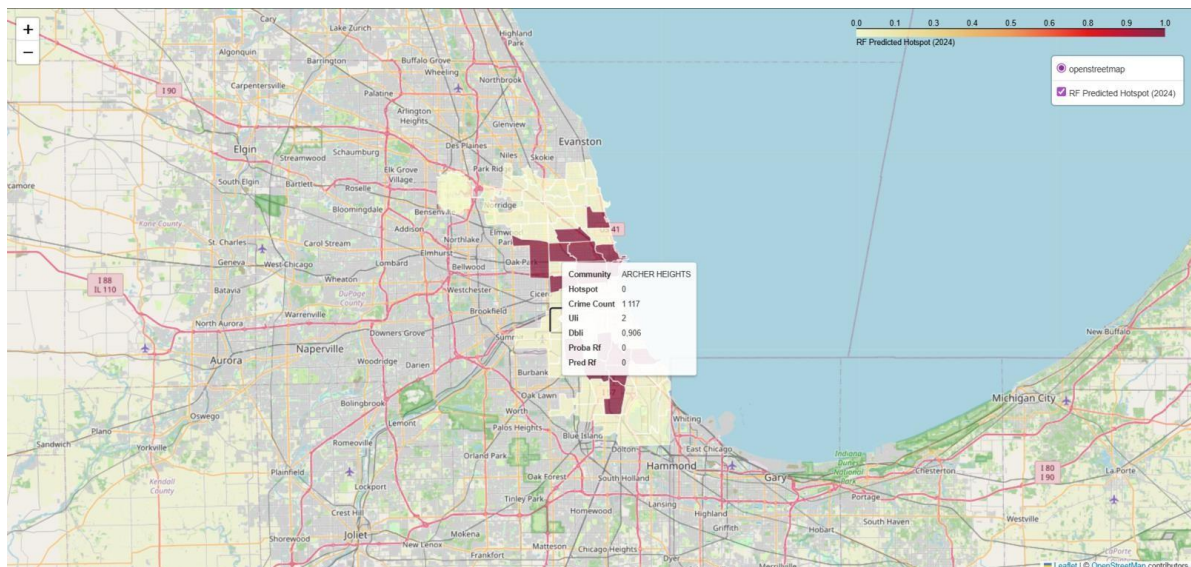
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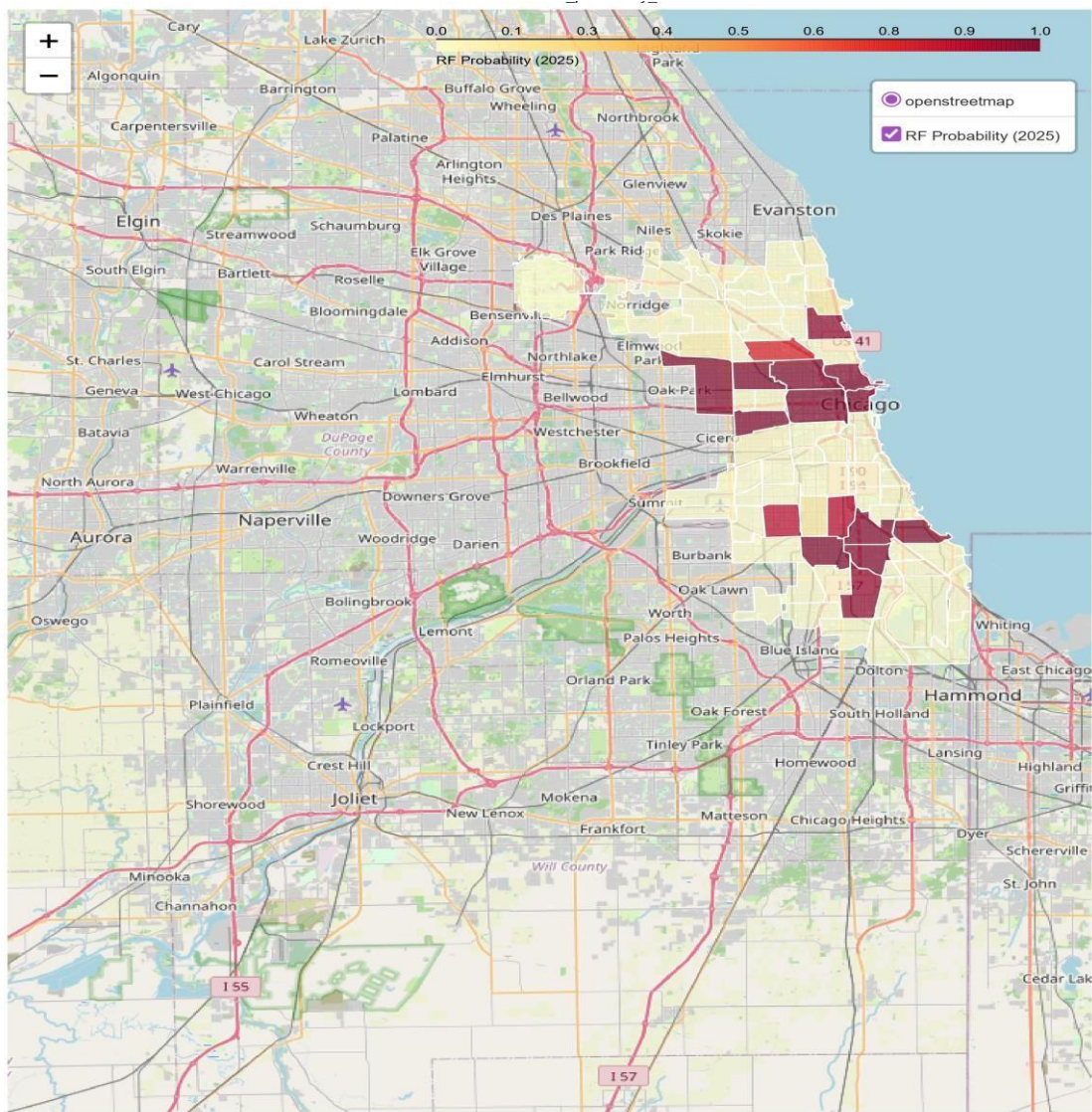




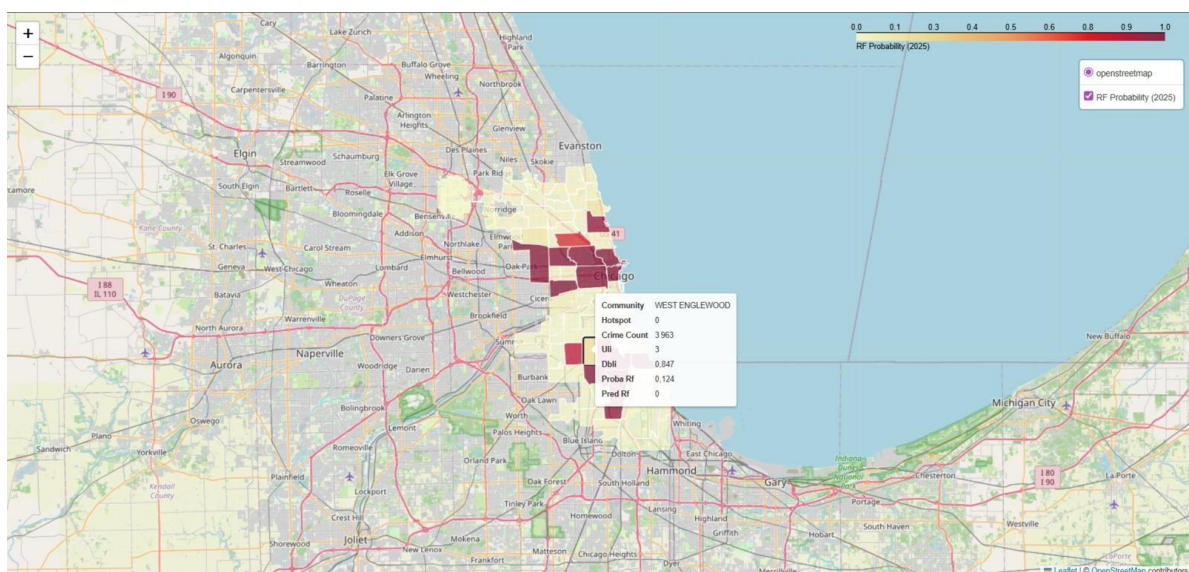


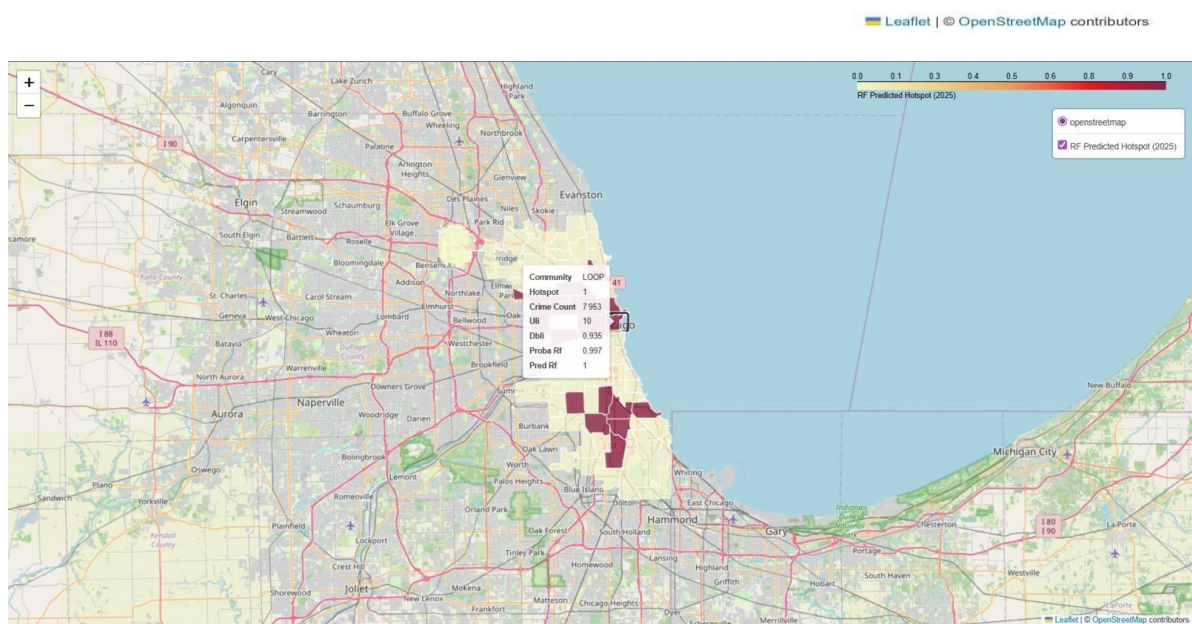
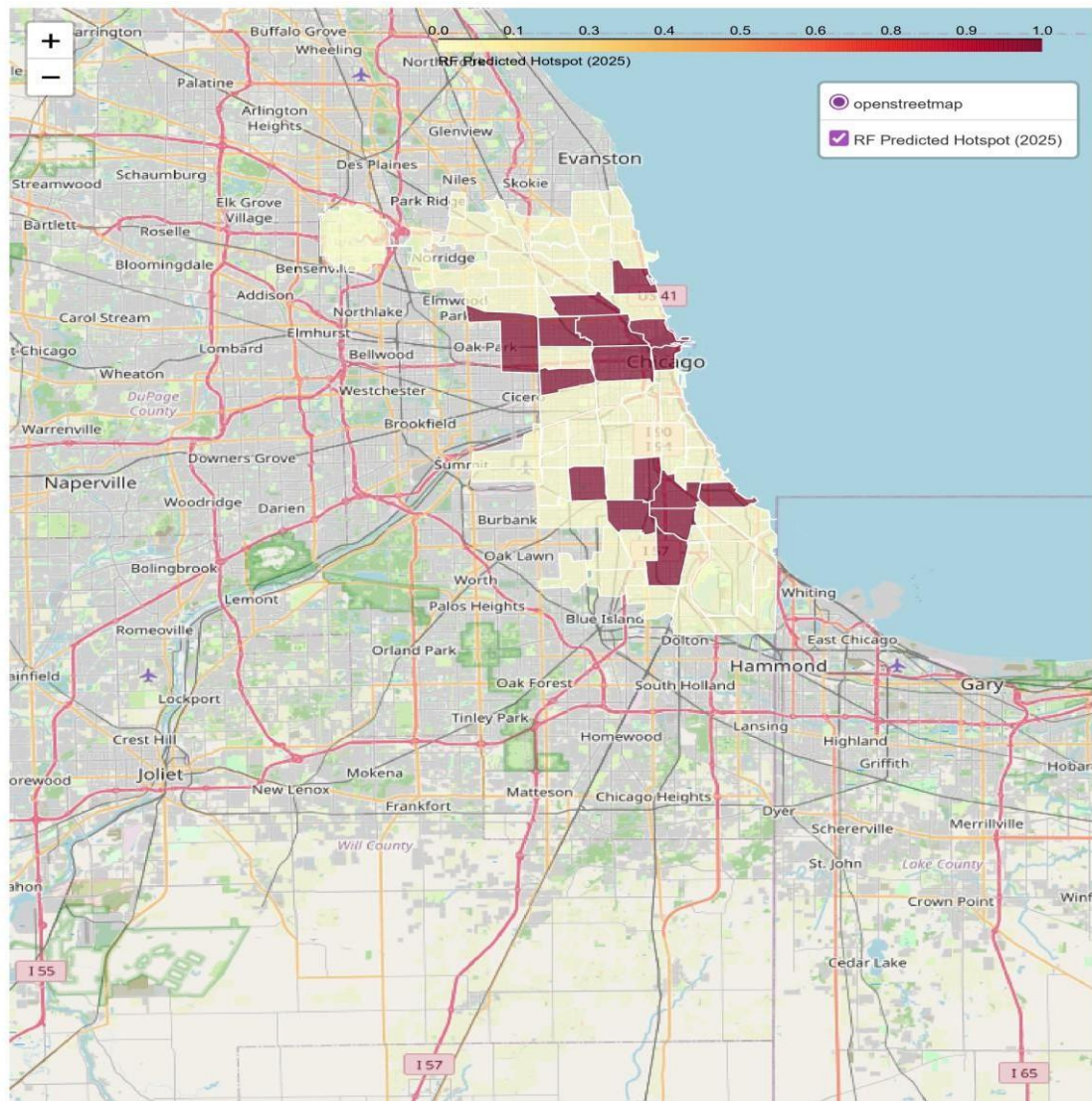
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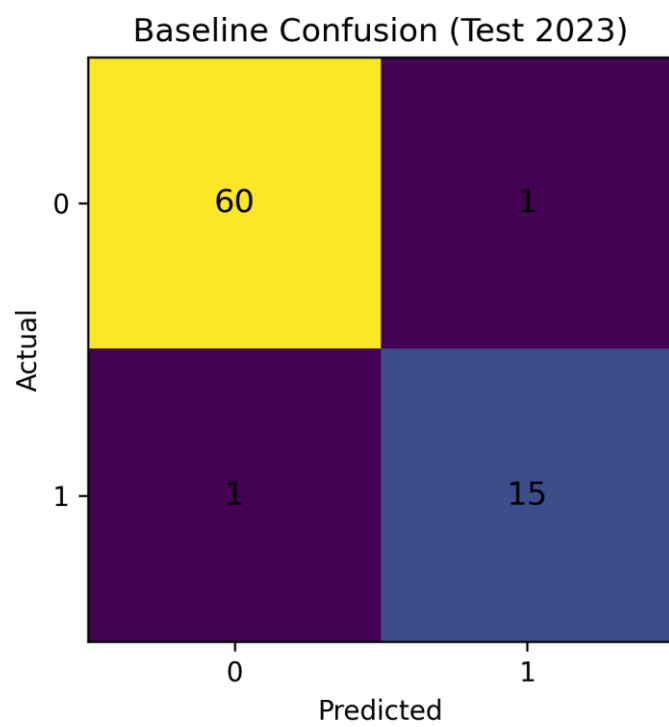
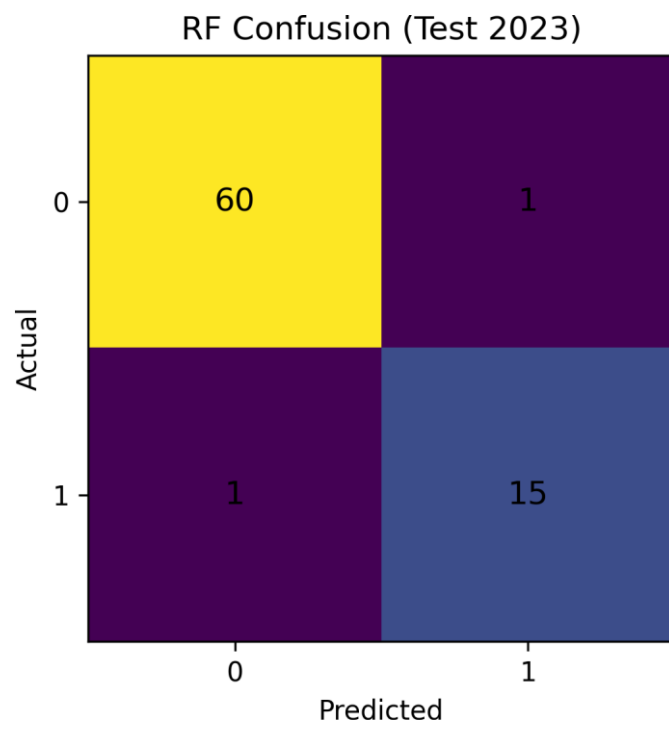


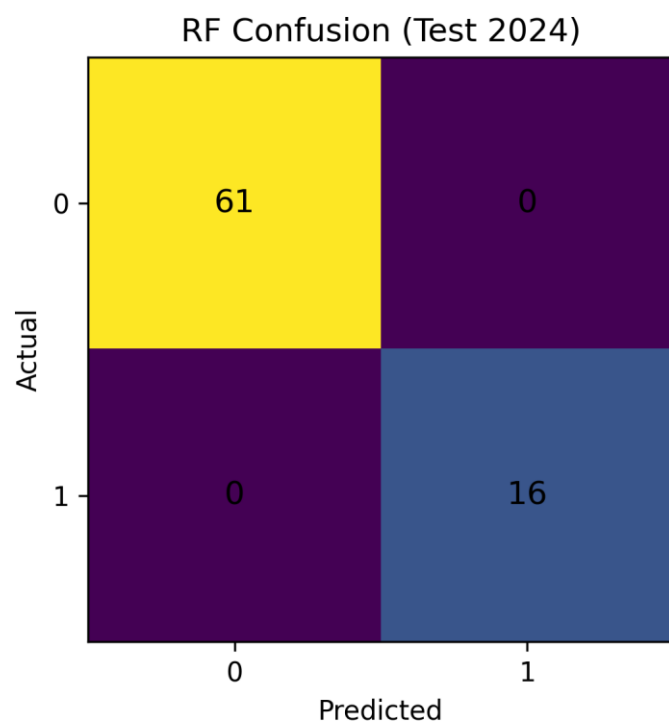
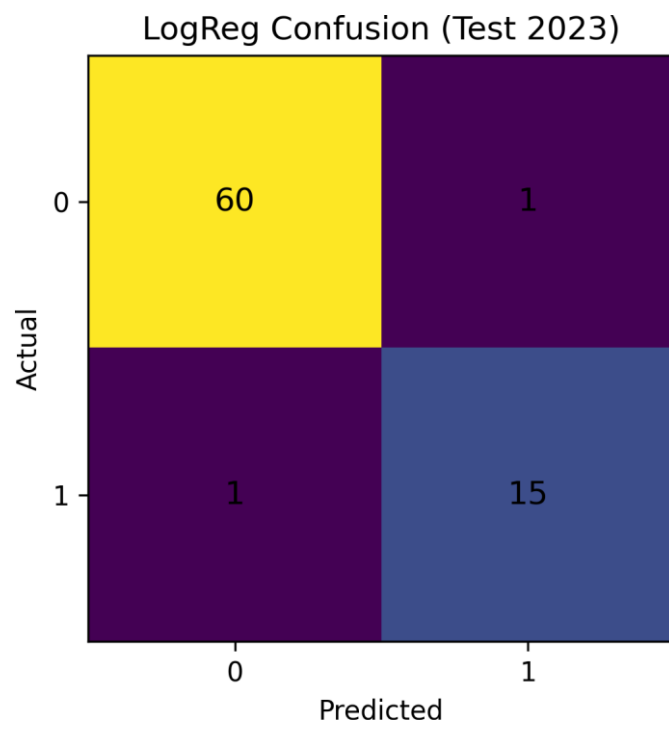


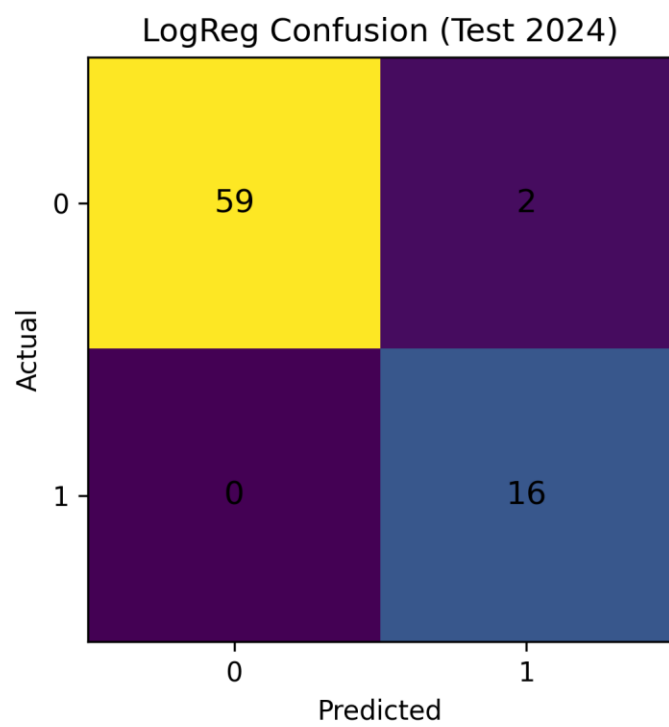
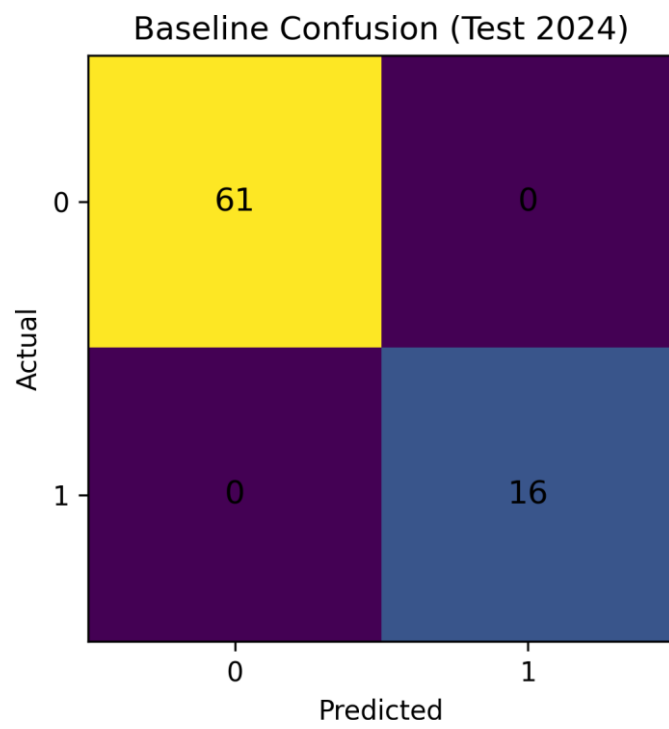
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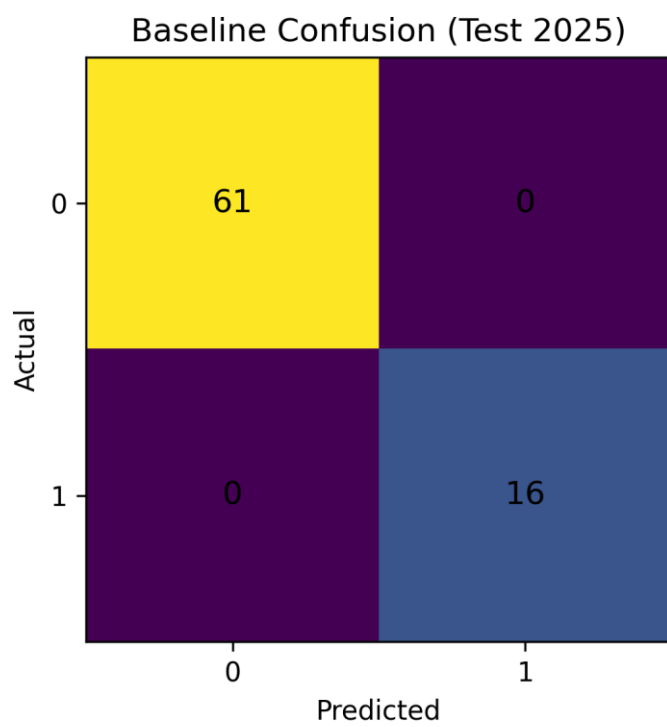
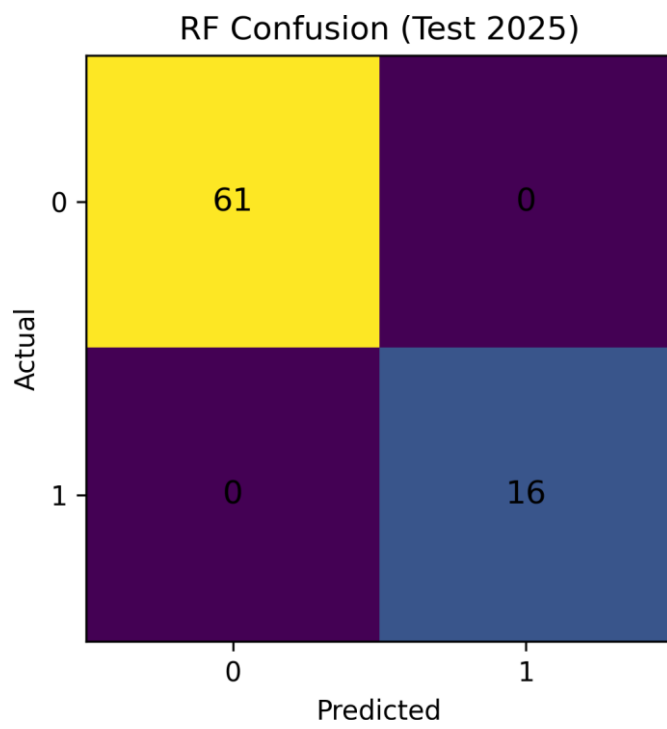


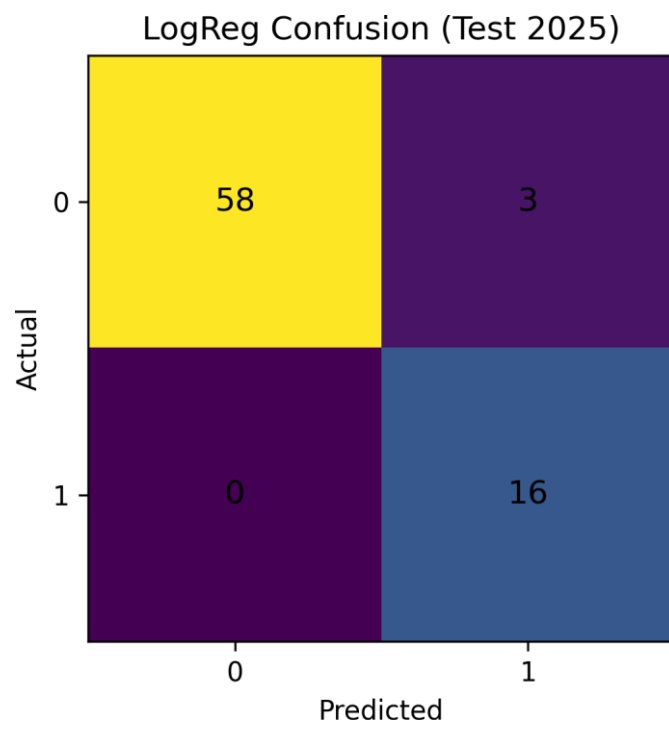












Rolling Backtest Performance by Year (Train $\leq$ y-1)

scheme	year	model	accuracy	precision	recall	f1	roc_auc	pr_auc
rolling_train $\leq$ y-1	2023	baseline_hotspot_lag1	0.974	0.9375	0.9375	0.9375	nan	nan
rolling_train $\leq$ y-1	2023	logistic_regression	0.974	0.9375	0.9375	0.9375	0.9939	0.9794
rolling_train $\leq$ y-1	2023	random_forest	0.974	0.9375	0.9375	0.9375	0.959	0.9331
rolling_train $\leq$ y-1	2024	baseline_hotspot_lag1	1.0	1.0	1.0	1.0	nan	nan
rolling_train $\leq$ y-1	2024	logistic_regression	0.961	0.8421	1.0	0.9143	1.0	1.0
rolling_train $\leq$ y-1	2024	random_forest	1.0	1.0	1.0	1.0	1.0	1.0
rolling_train $\leq$ y-1	2025	baseline_hotspot_lag1	1.0	1.0	1.0	1.0	nan	nan
rolling_train $\leq$ y-1	2025	logistic_regression	0.987	0.9412	1.0	0.9697	1.0	1.0
rolling_train $\leq$ y-1	2025	random_forest	1.0	1.0	1.0	1.0	1.0	1.0

Model Performance by Year (Fixed Train $\leq$ 2022)

scheme	year	model	accuracy	precision	recall	f1	roc_auc	pr_auc
fixed_train $\leq$ 2022	2023	baseline_hotspot_lag1	0.974	0.9375	0.9375	0.9375	nan	nan
fixed_train $\leq$ 2022	2023	logistic_regression	0.974	0.9375	0.9375	0.9375	0.9939	0.9794
fixed_train $\leq$ 2022	2023	random_forest	0.974	0.9375	0.9375	0.9375	0.959	0.9331
fixed_train $\leq$ 2022	2024	baseline_hotspot_lag1	1.0	1.0	1.0	1.0	nan	nan
fixed_train $\leq$ 2022	2024	logistic_regression	0.974	0.8889	1.0	0.9412	1.0	1.0
fixed_train $\leq$ 2022	2024	random_forest	1.0	1.0	1.0	1.0	1.0	1.0
fixed_train $\leq$ 2022	2025	baseline_hotspot_lag1	1.0	1.0	1.0	1.0	nan	nan
fixed_train $\leq$ 2022	2025	logistic_regression	0.961	0.8421	1.0	0.9143	1.0	1.0
fixed_train $\leq$ 2022	2025	random_forest	1.0	1.0	1.0	1.0	1.0	1.0

Random Forest Feature Importance (trained $\leq$ 2024)

